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Faculty of Business and Economics

Research Group Digital Health

Design and Implementation of a RAG-Enhanced Habit Recommendation System: A Design Science Research Study

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Submitted by: Jingting Hua

Student Number: 4910816

Supervisor: Dipl.-Wi.-Ing. Felix Reinsch

Marcel Susky M.Sc.

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Abstract

The modifiable factors of noncommunicable diseases (NCDs) are closely related to everyday habits. Although AI-supported Health Recommender Systems (HRSs) can support behavior change, existing research still lacks reusable design knowledge for integrating habits and their contextual information, NCD-related knowledge, user profiles, and feedback into interpretable and trustworthy recommendation processes.

Following the Design Science Research Methodology (DSRM), this research first identifies problems in the field of AI-supported HRSs, then derives meta-requirements and translates them into design principles and design decisions. Based on these design decisions, a Retrieval-Augmented Generation (RAG)-enhanced and ontology-informed Habit Recommender System was designed and implemented. The system was then evaluated through a preliminary persona-based evaluation using three literature-informed personas. The results indicate that the main technical workflows could be successfully executed in the evaluated persona-based episodes. However, the system did not always generate recommendations that fully reflected user-specific style preferences, and in a small number of cases, unsupported concrete numerical details were generated.

This thesis contributes a traceable design knowledge structure for AI-supported HRSs by linking literature-informed problem identification to meta-requirements, design principles, design decisions, and a working artifact.

Contents

1	Introduction	1
2	Basics and Related Research	5
2.1	AI-supported HRSs: A Non-systematic Literature Review	5
2.2	Opportunities and Risks of LLMs in RSs	11
2.3	RAG	12
2.4	Habit Modeling (Summary of the Systematic Literature Review)	13
3	Research Design	17
4	Habit Recommender System	21
4.1	Identify Problem & Motivate	21
4.2	Define Objectives of a Solution	25
4.3	Design & Development	28
4.3.1	Derivation of DPs from Meta-Requirements	29
4.3.2	Derivation of Traceable DDs	31
4.3.3	Conceptual Design	45
4.3.4	Modular Structure of the Prototype	47
4.3.4.1	Workflow 1	48
4.3.4.2	Workflow 2	49
4.3.4.3	Workflow 3	50
4.3.5	Prototype Implementation	52
4.4	Demonstration	57
4.5	Evaluation	63
5	Discussion	72
6	Conclusion and Future Work	73
	List of Figures	I
	List of Tables	II
	List of Listings	III
	References	IV

1 Introduction

Noncommunicable diseases (NCDs), also known as chronic diseases, refer to a group of conditions characterized by a long duration and slow progression. They primarily include four major categories: cardiovascular diseases (CVD), cancer, chronic respiratory diseases (CRD), and diabetes ([Budreviciute et al., 2020](#)). In terms of the global impact of noncommunicable diseases, NCDs killed at least 43 million people in 2021, equivalent to 75% of non-pandemic-related deaths globally. Among NCD-related deaths in 2021, cardiovascular diseases accounted for the largest share, causing at least 19 million deaths, followed by cancer (10 million), chronic respiratory diseases (4 million), and diabetes (over 2 million, including deaths from diabetes-related kidney disease). In terms of affected countries, 73% of these deaths occurred in low- and middle-income countries ([World Health Organization, 2025b](#)). With the expansion of globalization, rapid urbanization, and the prevalence of sedentary lifestyles, NCDs have gradually become a global issue, significantly impacting both developed and developing countries and imposing an increasingly severe social and economic burden ([Abegunde et al., 2007](#)).

Therefore, strengthening the prevention and control of NCDs has become an urgent priority. To effectively prevent and control NCDs, it is necessary to focus on the risk factors for NCDs—that is, factors that increase the likelihood of developing NCDs ([Budreviciute et al., 2020](#)).

According to a classification framework proposed by relevant studies, risk factors for NCDs can be categorized into five major groups: genetic factors, environmental factors, sociodemographic factors, factors related to self-management, and factors related to medical conditions. Among these, genetic, environmental, and sociodemographic factors are largely stable background determinants that individuals often find difficult to directly alter in the short term. In contrast, self-management factors and medical condition factors are more closely tied to modifiable lifestyle choices and thus represent key entry points for the prevention and control of NCDs. Among self-management factors in particular, unhealthy dietary habits, physical inactivity, smoking, and excessive alcohol consumption are typically regarded as classic modifiable risk factors ([Budreviciute et al., 2020](#)). At the same time, research indicates that many NCD risk factors can be traced back to early life and continue to accumulate and amplify throughout subsequent development. Therefore, from a life-course perspective, maintaining healthy lifestyles and behavioral habits throughout the life span contributes to the prevention and management of NCDs ([Budreviciute et al., 2020](#); [Kelishadi, 2022](#)).

In practice, these modifiable risk factors often correspond to specific daily behavioral patterns, such as dietary habits ([Adriaanse et al., 2011](#)), physical activity ([Rhodes and de Bruijn, 2013](#)), smoking, and alcohol consumption ([Norman, 2011](#)). These behaviors typically exhibit the characteristics of habits: they are repeated frequently in stable contexts (such as fixed times,

locations, preceding behaviors, or the presence of other people), thereby gradually forming “context–response associations”: This means that once a habit is established, contextual cues can automatically trigger impulses even in the absence of explicit intention, and the behavior is relatively “insensitive” to short-term goal changes or rational deliberation (Wood and Neal, 2007; Wood and R  nger, 2016). A common example is as follows: “I get home around 7:30 p.m. every day, put down my backpack, sit on the living room sofa, and watch TV while snacking.” High-frequency repetition in a stable context causes the brain to associate the behavioral response of “eating snacks” with the sequence “arriving home at 7:30 p.m. + sitting on the sofa + turning on the TV,” or even with the entire situational cue. Even if there is a short-term goal, such as “I need to lose weight lately,” one may still find oneself opening a bag of snacks and starting to eat.

Therefore, for forms of risk behavior that are shaped by contextual cues, general and occasional health education alone is often insufficient to achieve lasting effects in everyday life. Instead, individuals need actionable recommendations in relevant daily situations to help them disrupt unhealthy habits and establish or maintain healthier behaviors (Wood and R  nger, 2016). In the field of information systems, the task of delivering actionable and personalized suggestions to users is commonly conceptualized as a recommender system (RS). RSs are generally understood as software-based approaches that generate suggestions for items that may be useful or relevant to a user (Ricci et al., 2011). Such systems have been widely used in domains including e-commerce, content platforms such as news, video, and music, social networks, local and mobility services, recruitment, and education, where they support users in selecting relevant options from a large set of alternatives (Ricci et al., 2011). When these approaches are applied in health-related contexts, they are commonly referred to as Health Recommender Systems (HRSs). In this context, HRSs can be understood as digital systems that support health management by providing personalized recommendations based on users’ health history, behavior, and preferences, thereby helping them make better-informed health decisions (Barbaric et al., 2025). Accordingly, RSs designed for the prevention and management of NCDs can be regarded as a type of HRS.

Recommendation algorithms are a core component of RSs and can vary significantly in nature. Major variants include collaborative filtering (CF) RSs (Schafer et al., 2007), content-based RSs (Sanchez Bocanegra et al., 2017), and hybrid RSs (Adomavicius and Tuzhilin, 2005). The core idea of CF is to mine similarities from group interaction data (such as ratings, clicks, purchases, and adoptions). If two users exhibit similar preferences in their historical interactions, they are likely to be interested in similar items. Similarly, if two items are preferred by a large number of similar users, they are likely to appeal to the same type of users (Schafer et al., 2007). In contrast, content-based RSs take a different approach; they do not rely primarily on user interactions

but focus on “what the item itself is.” They characterize items using their content and attribute features (such as text descriptions, tags, and categories), summarize preference profiles from items the user has previously liked or adopted, and then recommend items most similar to that profile (Sanchez Bocanegra et al., 2017). Furthermore, hybrid RSs combine CF with content features to leverage the strengths of both approaches (Adomavicius and Tuzhilin, 2005).

Since CF relies on user–item interactions to calculate similarity, sparsity is a common problem in CF systems, as users typically rate or interact with only a small fraction of items. As a result, there is insufficient common ground, making it difficult to establish reliable relationships between similar users and similar items (Ricci et al., 2011). At the same time, CF systems also face a cold-start problem: new users or items may have no ratings at all. In particular, new items may be impossible to recommend when no one has rated them. In contrast, content-based RSs rely on the items’ own features, making them easier to implement in new-item scenarios. However, they also need to learn preferences from user history in order to build user profiles. Therefore, they likewise face a cold-start problem for new users (Ricci et al., 2011).

To overcome challenges such as sparsity and the cold-start problem, research typically introduces more sophisticated mechanisms. Artificial intelligence (AI) has gradually become a crucial technological foundation for RSs, helping them learn predictive models from data and improve recommendation quality (Lopez-Barreiro et al., 2024a). AI can generally be understood as the science and engineering of creating intelligent machines and computer programs (McCarthy, 2007). In recent years, with the development of large language models (LLMs), these models have also been applied to RSs (Wu et al., 2024). LLMs represent one important class of AI models. Typically, LLMs refer to large-scale Transformer-based language models (Zhao et al., 2025). The key advantages of integrating LLMs into RSs lie in their ability to extract high-quality representations of textual information and leverage the vast amount of external knowledge encoded in their parameters, thereby improving the accuracy and relevance of recommendations and enhancing user satisfaction. At the same time, LLMs open up new possibilities for RSs through their zero-shot and few-shot recommendation capabilities (Wu et al., 2024; Zhao et al., 2024).

However, research on the design of AI-supported HRSs remains limited (Wu et al., 2024; Zhao et al., 2024; Wang et al., 2024). In particular, the field still lacks a reusable, context-driven design knowledge framework, which limits the generalizability of existing approaches. Most AI-supported HRSs are currently tailored to specific domains, such as diet- or exercise-focused systems, and differ in the extent to which they address the prevention of NCDs (Yang et al., 2024; Galitsky, 2024; Hemaraju et al., 2023). Even when such systems perform well in one or a few domains, they often struggle to provide sustainable and holistic intervention support that is

adaptable to real-life contexts. In other words, there is still a lack of cross-domain and scalable design knowledge for AI-supported HRSs. This gap includes the absence of meta-requirements for AI-supported HRSs, as well as a lack of guidance on how such meta-requirements can be systematically translated into actionable design principles (DPs) and further operationalized in specific design decisions (DDs) for artifacts ([Kopenhagen et al., 2012](#)).

This thesis reports a comprehensive Design Science Research (DSR) project that developed and evaluated a habit recommendation system, hereafter referred to as the Habit Recommender System, integrating Retrieval-Augmented Generation (RAG), multiple LLMs, and ontology-based knowledge to support behavior change in users' everyday lives and contribute to the prevention of NCDs. Methodologically, the study adopts the DSR paradigm and follows the Design Science Research Methodology (DSRM) process model proposed by [Peffer et al. \(2007\)](#). With regard to research contributions, the study not only produces the corresponding artifact, but also systematically identifies meta-requirements and derives DPs and DDs for this domain. These outcomes provide a foundation for the future design and evaluation of AI-supported HRSs. Based on this, the following research questions are proposed:

RQ1: What meta-requirements should AI-supported HRSs fulfill?

RQ2: Based on these meta-requirements, which transferable DPs can be derived?

RQ3: Based on these DPs, which transferable DDs can be derived, and how can they be translated into a corresponding AI-supported HRS artifact?

RQ4: To what extent does the resulting artifact satisfy the identified meta-requirements?

The remainder of this thesis is structured as follows. The next section presents the basics and related research. It introduces the conceptual and technical foundations of the thesis, including AI-supported HRSs, LLMs in recommender systems, RAG, and habit modeling. The following section describes the research design and explains how the DSR paradigm and the DSRM process model are applied in this thesis. The subsequent section presents the Habit Recommender System following the main DSRM activities, including problem identification, the derivation of meta-requirements, DPs and DDs, artifact design and implementation, demonstration, and preliminary evaluation. The thesis then discusses the main findings and limitations of the research. Finally, it concludes by summarizing the answers to the research questions and outlining directions for future work.

2 Basics and Related Research

This chapter provides the theoretical and technical foundation for the present study. First, the subsection AI-supported HRSs: A Non-systematic Literature Review reviews the existing literature on AI-supported HRSs through a non-systematic literature review in order to obtain a structured overview of the current research landscape. In this context, the reviewed studies are examined from multiple perspectives, including domain coverage, target population, input data, the role of ontologies, the role of NCDs, feedback mechanisms, interpretability, the use of LLMs, and technical challenges such as cold-start and sparsity problems. The subsection Opportunities and Risks of LLMs in RSs then introduces LLMs and discusses the roles they have played in RSs so far, as well as the main opportunities and risks associated with their use. This is followed by an introduction to RAG, including its basic concept, its role in RSs, its architecture, and its recent developments. Finally, this chapter turns to habit modeling by outlining the concepts of habits and contextual cues, explaining the relationship between habits and behavior change, and summarizing the main findings of the author's previous systematic literature review on habit modeling methods conducted in a research seminar (Forschungsseminar). Taken together, these perspectives establish the conceptual foundation for the design of the proposed AI-supported HRS artifact.

2.1 AI-supported HRSs: A Non-systematic Literature Review

To better understand the current state of research on AI-supported HRSs, a non-systematic literature review was conducted. The search string consists of four main components: (1) the review focuses on RSs; (2) it focuses on RSs in the health field, i.e., HRSs; (3) it focuses only on AI-supported HRSs; and (4) the primary objective of the system is to promote behavior change.

$$(\text{"recommender system"} \vee \text{"recommendation system"}) \quad (1)$$

$\wedge$

$$(\text{health} \vee \text{"digital health"} \vee \text{"health care"}) \quad (2)$$

$\wedge$

$$(AI \vee \text{"artificial intelligence"}) \quad (3)$$

$$\wedge$$

$$(\text{"behavior* change"} \vee \text{"habit formation"}) \quad (4)$$

Using the defined search string, a search was conducted in Scopus, EBSCO (All 31 databases), Web of Science (Web of Science Core Collection), PubMed, ACM Library (ACM Full-Text Collection), and AIS eLibrary (AIS Electronic Library). The results were deduplicated, and English-language articles matching the Search String were selected. Review articles were excluded, but backward and forward snowballing were conducted based on these review articles (Wohlin, 2014). The search yielded 36 relevant articles, and the extracted domains and recommended objects are presented in Table 1.

Table 1: Summary of Related Studies on AI-Supported HRSs

Author/Year	Domain	Object
(Mojarad et al., 2020a)	Activities of daily living	Well-being activities
(Afonso et al., 2020)	Childhood obesity: physical activity and diet and sleep	Educational videos related to eating, drinking, moving, and sleeping habits
(Hammer et al., 2010)	Diabetes management: physical activity and diet	Meal plans and physical activity
(Yang et al., 2024)	Diet	Food items
(Xu and Wang, 2025)	Diet	Healthy food
(Galitsky, 2024)	Diet	Related to diagnoses, health insurance, treatment methods
(Zhang et al., 2025)	Diet	Food items and healthiness interpretation
(Nouh et al., 2019)	Diet	Healthy diet menu items and alternative diet menu suggestions
(Orte et al., 2023)	Diet	Personalized diet missions and supporting recommendation content
(Hamdollahi Oskouei and Hashemzadeh, 2023)	Diet	recipes and meal/snack calorie-cap constraints

Table 1: Continued.

Author/Year	Domain	Object
(Silva et al., 2022)	Diet	Healthy diet-group list
(Ramesh et al., 2021)	Indian elderly diabetic population: diet	Food items list
(Ramaraj et al., 2024)	Indian female athletes and active adolescents: physical exercise and diet	Indian meal items and per-meal nutrient content
(Annapoorna et al., 2023)	Obesity management: physical exercise and diet	A diet plan and an exercise plan
(Costa et al., 2018)	Older adults: home-based physical exercise	Physical exercise activities
(Li et al., 2019)	Physical activity	An hour-by-hour activity plan, plus notifications and suggested plan adjustments
(Vairavasundaram et al., 2022)	Physical activity	An hourly step target plan for the rest of the day
(Chatterjee et al., 2022a)	Physical activity	Physical activity plan
(Ali et al., 2016)	Physical activity	weight status and management plan and actionable activity plan
(Zhao et al., 2020)	Physical activity	The type of physical activity
(Chatterjee et al., 2022b)	Physical activity	Goal oriented activity suggestions and reminder content
(Palomares et al., 2022)	Physical activity and diet	Meals and physical activity bundle
(Hemaraju et al., 2023)	Physical activity and diet	Meal items and physical activity types
(Vecchia et al., 2024)	Physical activity and sleep	Next-step and avoid activity types, and predicted sleep quality score.
(Erdeniz et al., 2020)	Physical activity and sleep	Activity/sleep plans and IoT device
(Chen et al., 2021a)	Physical exercise	Exercise prescriptions
(Tran et al., 2018)	Physical exercise	Workout plans and training parameters
(Wang et al., 2023)	Physical exercise and diet	Recipe-based meal plans (breakfast/lunch/dinner recipe bundles) and exercise items

Table 1: Continued.

Author/Year	Domain	Object
(Balpande et al., 2023)	Physical exercise and diet	Weekly training schedule/recommended exercise list and weekday meal plan
(Jamil et al., 2021b)	Physical exercise and diet	Diet plans and workout plans
(Donciu et al., 2011)	Physical exercise and diet	Diet plans and workout plans
(Jamil et al., 2021a)	Physical exercise and diet	Diet plans and training plans
(Mahyari and Pirolli, 2021)	Physical exercise and mental health	Next activity/goal and predicted success probability
(Chen et al., 2021b)	Smoking cessation	Motivational smoking cessation messages
(Sadashivam et al., 2016)	Smoking cessation	Motivational smoking cessation messages
(Hors-Fraile et al., 2019)	Smoking cessation	Motivational smoking cessation messages

In terms of domain coverage, the 36 reviewed articles mainly focused on a single domain, including diet (9 studies), physical activity (9 studies), smoking cessation (3 studies), and activities of daily living (1 study), or on two or more domains (14 studies). This suggests that AI-supported HRSs are increasingly oriented toward more comprehensive lifestyle intervention scenarios. However, there is still a lack of AI-supported HRSs that cover a broad range of domains or allow users to flexibly adjust the recommendation domain according to their individual needs.

In terms of the target population, a few studies (4 studies) focused on specific groups, such as children with obesity (Afonso et al., 2020), people with diabetes (Hammer et al., 2010), older adults (Ramesh et al., 2021; Costa et al., 2018), and female athletes and adolescents (Ramaraj et al., 2024).

In terms of input data, most of the 36 studies relied on multi-source heterogeneous data to support personalized health recommendations, mainly covering five categories. First, user-explicit input data, such as basic information, health goals, preferences, and questionnaire feedback. Second, dynamically collected data passively captured by the system, such as step counts, sleep, activity intensity, and physiological indicators recorded by wearable devices and smartphone sensors. Third, medical and health status data, such as medical history, clinical

test results, and Personal Health Record information. Fourth, contextual information, such as time, location, and weather. Fifth, external knowledge sources, such as food, exercise, and nutrition databases, as well as expert rules. In addition, some studies (5 studies) further employed ontologies to model the data ([Chatterjee et al., 2022a,b](#); [Mojarad et al., 2020a](#); [Donciu et al., 2011](#); [Hamdollahi Oskouei and Hashemzadeh, 2023](#)).

An ontology refers to a formal semantic structure for knowledge representation that is used to model, in a unified manner, the core concepts of a specific domain and the relationships among them, thereby providing a semantic foundation for subsequent information extraction, evidence annotation, automated reasoning, and the construction of knowledge systems ([Norris et al., 2017](#); [Taji and Lange, 2016](#)).

In terms of the role of ontologies, the existing studies can be grouped into three categories. The first category employs ontologies as a layer for semantic integration and behavioral context modelling. In these studies, ontologies are used to organize activity data, contextual information, user preferences, and related elements into a unified semantic representation, which is then combined with rules, reasoners, the SPARQL Protocol and RDF Query Language (SPARQL), or Probabilistic Answer Set Programming (PASP) to determine user states and generate personalized recommendations ([Chatterjee et al., 2022a,b, 2021](#); [Mojarad et al., 2020a](#)). The second category uses ontologies as semantic knowledge bases. In this case, the ontology mainly captures knowledge related to nutrition, sports medicine, and runners, and is queried through Sesame, SPARQL, and Jena to support the recommendation process ([Donciu et al., 2011](#)). The third category uses ontologies for feature organization and vectorization preparation. More specifically, food and user ontologies are used to systematically structure heterogeneous food- and user-related features, which are subsequently transformed into vector representations for downstream recommendation models ([Hamdollahi Oskouei and Hashemzadeh, 2023](#)).

In terms of how much attention is paid to NCDs, existing health recommender research can be broadly divided into three categories. Most studies treat NCDs primarily as overarching health goals or motivations for behavior change (i.e., as background), but do not incorporate users' chronic disease information into the recommendation logic. The second category (three studies) begins to incorporate users' chronic disease information into user profiles, allowing NCDs to participate in recommendation through forms such as self-descriptions and authorized documents; however, such information usually remains only one of many individual features rather than a core constraint in the recommendation process ([Xu and Wang, 2025](#); [Galitsky, 2024](#); [Hamdollahi Oskouei and Hashemzadeh, 2023](#)). The third category (three studies) places NCDs more centrally in the recommendation logic, meaning that the systems directly construct tags, rules, and knowledge bases around specific chronic diseases or related health needs ([Zhang](#)

et al., 2025; Ramesh et al., 2021; Hammer et al., 2010). For example, MOPI-HFRS generates health tags from dietary status and medical data, and maps conditions such as hypertension, low-salt diets, or renal diets to user tags such as `low_sodium` (Zhang et al., 2025).

Explicit feedback refers to direct observations of preferences, with typical forms including ratings, likes, and dislikes. Implicit feedback, by contrast, reflects users' preferences indirectly through the observation of user behavior, with common forms including purchases, browsing, search activities, and mouse movements. In both cases, such feedback should be able to influence subsequent recommendations (Hu et al., 2008; Rendle et al., 2012; Oard and Kim, 1998). Among the 36 studies reviewed, approximately two-thirds (23 studies) incorporated feedback mechanisms. Of these, 15 studies included only explicit feedback, 13 studies included only implicit feedback, and 5 studies combined both types of feedback. By contrast, approximately one-third (13 studies) did not demonstrate any feedback mechanism.

RSs that employ algorithms that are too complex to be understood by users, or that remain inaccessible due to proprietary constraints such as trade secrets, can be characterized as black-box recommenders (Rudin, 2019; Herlocker et al., 2000). Because such systems lack transparency and interpretability, users may find it more difficult to identify and correct recommendation errors, which can in turn reduce user acceptance of and trust in the system. Moreover, black-box decision-making systems have already raised concerns in high-stakes domains such as healthcare and criminal justice (Rudin, 2019; Herlocker et al., 2000).

Therefore, it is crucial for RSs to provide a meaningful chain of evidence for their recommendations. In terms of interpretability, only a small number of studies (4 studies) implemented explicit explanation mechanisms that clarify why a recommendation was made. These include LLM-based systems that combine users' health conditions with nutritional information to generate recommendation justifications (Yang et al., 2024; Zhang et al., 2025; Galitsky, 2024), as well as rule-based systems that generate explanations from explicit rules and directly present the conditions triggering the recommendation to the user (Vecchia et al., 2024).

In terms of the use of LLMs in recommendation systems, only a small number of studies (4 studies) incorporated LLMs, and these models played different roles across the systems. Compared with Yang et al. (2024); Zhang et al. (2025), which mainly used LLMs for recommendation explanation and user interaction, Galitsky (2024) went a step further by placing LLMs at the center of personalized recommendation. In that study, the models were used for tasks such as user profile summarization, prompt construction, recommendation generation based on integrated search results, and explanation generation. In addition, these studies generally pointed out that LLMs are prone to hallucinations when domain-specific or specialized knowledge is insufficient. To mitigate this issue, prior work mainly relied on prompt engineering or prompt

learning to constrain the generation process, or introduced search engines and recommendation engines to provide external knowledge support (Yang et al., 2024; Zhang et al., 2025; Galitsky, 2024). LLMs can also be used to generate semantically richer embeddings, which are then incorporated into recommendation systems (Xu and Wang, 2025).

In terms of cold-start and sparsity problems, 8 studies explicitly stated that they encountered one or both of these issues. Cold start usually refers to the lack of available historical information (e.g., new users, new items, or new health contexts), which prevents the system from generating reliable recommendations. Sparsity, by contrast, mainly refers to the large number of missing values in user–item rating matrices, feedback matrices, or case/device datasets, which leads to declines in similarity calculation, preference learning, or prediction accuracy. When a system relies on collaborative filtering or content-based filtering algorithms, it naturally gives rise to cold-start and sparsity problems to a certain extent. The reason is that such methods usually infer individual preferences based on existing historical information. Therefore, once users lack sufficient historical data, the system finds it difficult to build stable preference representations, thereby giving rise to the typical cold-start problem. At the same time, because in real-world scenarios most users provide only limited feedback on a small number of items, user–item matrices often contain a large number of unobserved values, which in turn results in data sparsity. Therefore, in recommendation frameworks centered on similarity-based modeling, cold start and sparsity are not incidental problems but are closely related to the algorithmic mechanism itself (Chen et al., 2021b; Sadasivam et al., 2016; Hors-Fraile et al., 2019; Nouh et al., 2019; Erdeniz et al., 2020; Donciu et al., 2011). In addition to traditional collaborative filtering and content-based filtering, some models also rely on a certain amount of individual historical data for personalized modeling. Therefore, when user histories are too short, sample sizes are too small, or sufficient prior records are lacking, system performance also declines significantly (Mahyari and Pirolli, 2021; Xu and Wang, 2025).

2.2 Opportunities and Risks of LLMs in RSs

This thesis defines LLMs as a class of models based on the Transformer architecture, characterized by a large number of parameters and trained on large-scale corpora using self-supervised or semi-supervised learning, such as Bidirectional Encoder Representations from Transformers (BERT) and Generative Pre-trained Transformer (GPT) models (Wu et al., 2024; Zhao et al., 2024). The Transformer replaces traditional recurrent and convolutional structures with a self-attention mechanism, thereby significantly improving parallel training efficiency (Vaswani et al., 2017). Over time, LLMs have demonstrated strong capabilities in contextual understanding, text generation, and task transfer, which has led to their increasing adoption in recommender system

research ([Wu et al., 2024](#); [Zhao et al., 2024](#)).

In RSs, LLMs do not play a single role. According to the modeling paradigms proposed by [Wu et al. \(2024\)](#), existing studies can be broadly divided into three categories. The first category is LLM Embeddings + RS, in which LLMs are used as feature encoders to transform textual information about users and items into semantic representations, which are then combined with traditional recommendation models. The second category is LLM Tokens + RS, in which LLMs generate text tokens that reflect user preferences or item semantics, and these tokens are subsequently fed into downstream recommendation modules. The third category is LLM as RS, in which LLMs are more directly used as recommenders themselves by organizing user profiles, historical behaviors, candidate items, and task instructions into prompts to generate recommendation results directly. It should be noted that this classification emphasizes the functional role of LLMs in the recommendation process. Other review studies have also categorized this field from perspectives such as adaptation methods or system development frameworks, including pre-training, fine-tuning, and prompting ([Zhao et al., 2024](#)), as well as Representing and Understanding, Scheming and Utilizing, and Industrial Deploying ([Wang et al., 2024](#)).

Although LLMs have brought new capabilities to recommendation systems, they have also introduced a series of issues, among which hallucination is one of the most prominent. Hallucination refers to the model generating content that appears reasonable and coherent but is, in fact, unfounded, inaccurate, or even fabricated ([Rawte et al., 2023](#)). In recommendation scenarios, this issue may manifest as fabricating information about non-existent items, generating recommendations that cannot be mapped to the candidate pool, or providing explanations for recommendations that lack factual support. In high-stakes recommendation scenarios such as healthcare and legal contexts, such errors may further undermine system credibility and user safety ([Zhao et al., 2024](#); [Wu et al., 2024](#)).

2.3 RAG

The core idea of RAG technology is to combine the model’s internally learned “parametric knowledge” (i.e., the vague “memory” encoded in the model weights) with “non-parametric knowledge” from external knowledge bases (i.e., precise external data that can be updated at any time) ([Lewis et al., 2021](#); [Gao et al., 2024a](#); [Liu, 2025](#)).

RAG’s most important value lies not only in its ability to fill knowledge gaps in specialized domains by introducing externally retrievable knowledge, but also in its ability to reduce hallucinations by grounding the generation process in retrieved evidence ([Lewis et al., 2021](#); [Lála et al., 2023](#)). Compared with purely parametric generation baselines, RAG-generated

content also performs better in terms of specificity, diversity, and factuality (Lewis et al., 2021). More importantly, RAG can provide source grounding and traceability for its answers. This evidence-based characteristic is particularly important in high-stakes domains such as law and healthcare and helps enhance the credibility of system outputs (Lála et al., 2023; Ongri et al., 2025; Ning et al., 2025).

Its architecture typically consists of two stages. In the retrieval stage, the system uses an embedding model to transform the textual content of an external knowledge base into vector representations and constructs a corresponding vector index stored in a vector database, thereby enabling the structured organization of external knowledge (Mikolov et al., 2013; Gao et al., 2024a). When a user submits a query, the retrieval module employs the same embedding model to encode the question into a vector and performs similarity search over the large-scale knowledge base to retrieve the document chunks that are semantically most relevant to the query. In the generation stage, the retrieved document chunks and the user's original question are jointly fed into the generation module. Guided by a predefined prompt, the system integrates the contextual information with the question and drives the LLMs to produce answers that are more controllable, better grounded, and more consistent with external knowledge (Lewis et al., 2021; Gao et al., 2024a; Liu, 2025).

The development of RAG can generally be divided into three forms: Naive RAG, Advanced RAG, and Modular RAG (Gao et al., 2024a). Naive RAG usually follows a relatively basic linear “retrieve-then-generate” pipeline. Its core idea can be traced back to the original RAG framework, which combines external non-parametric memory with a parametric generation model to enable knowledge-augmented generation (Lewis et al., 2021). Advanced RAG further extends this framework by introducing optimization steps before and after retrieval, such as query rewriting and reranking, in order to improve retrieval quality and the final generation performance (Gao et al., 2024a). In contrast, Modular RAG places greater emphasis on designing the system as an orchestratable and composable modular architecture. Through mechanisms such as routing, query transformation, and fusion, different components can be flexibly adjusted and combined according to task requirements, thereby significantly improving the adaptability and scalability of the system (Gao et al., 2024b).

2.4 Habit Modeling (Summary of the Systematic Literature Review)

Modifiable risk factors for NCDs are closely linked to specific patterns of everyday behavior, such as eating habits (Adriaanse et al., 2011), physical activity (Rhodes and de Bruijn, 2013), smoking, and alcohol consumption (Norman, 2011). Therefore, a recommender system should collect users' habits in a timely manner to support the prevention and management of NCDs.

In this thesis, habits are defined as behaviors that are repeated frequently in stable contexts (e.g., fixed times, locations, preceding behaviors, or the presence of other people), thereby gradually forming context–response associations. This means that once a habit has been established, relevant contextual cues can automatically trigger an impulse even in the absence of explicit intention, and the behavior becomes relatively insensitive to short-term changes in goals or rational deliberation (Wood and Neal, 2007; Wood and R nger, 2016). A context–response association refers to the process of linking a specific context with a specific behavioral response stored in memory. Contextual cues that trigger habitual behavior may include the physical environment, other people, and preceding actions in a behavioral sequence (Wood and R nger, 2016). For example, getting up in the morning and walking into the kitchen may become associated with the behavior of drinking coffee. Morning can serve as a temporal contextual cue, getting up is a preceding action in the behavioral sequence, and the kitchen is the physical environment.

Habits are closely related to behavior change. Habit formation is achieved by repeatedly performing a certain behavior in a specific environment through cue-based learning until a context–response association is established (Wood and R nger, 2016). This principle generally applies to the formation of new healthy habits in a new environment (Carden and Wood, 2018). Once a context–response association has been established, contextual cues can trigger the automatic enactment of the habit. This mechanism has also been used to break undesirable habits; for example, changing the environment can change the habit (Wood and R nger, 2016; Wood and Neal, 2007). By deepening the understanding of habit mechanisms and implementing interventions aimed at changing lifestyle behaviors, it may be possible to successfully break these undesirable habits and thereby improve the modifiable risk factors of NCDs (Wood and R nger, 2016).

It is worth noting that the context of habits has not been specifically defined or systematically classified. Wood and Neal (2007) only mention that physical settings and preceding actions can serve as the context of habits. Wood (2017) proposes that the context of habits includes physical locations, other people, internal states, and preceding actions in a sequence, but it does not further explain the definition of each context category. At the same time, there is still no cross-domain, universally accepted classification method for modeling habits. To address this issue, in a previous research seminar, the author conducted a systematic literature review in accordance with the PRISMA 2020 guidelines (Page et al., 2021). Table 2 summarizes the modeled objects and main approaches related to habits and behavior identified in the review.

Table 2: Overview of Included Studies, Modeled Objects, and Main Approaches

Reference	Modeled Object	Main Approach
(Mojarad et al., 2020b)	Human Activity	Human Activity Ontology
(Kim et al., 2019)	Physical Activity	Physical Activity Ontology
(Khanfar et al., 2023)	Driving Behavior	K-Means Clustering Algorithm with Elbow Method
(Taji and Lange, 2016)	Human Food Habits and Eating Processes	The uc_Eating Ontology
(Sun et al., 2017)	Driving Behavior	Minimum Distance Classification Method
(Millonig and Gartner, 2008)	Spatio-Temporal Behaviour	Clustering Algorithms (Hierarchical Clustering, K-Means)
(Norris et al., 2017)	Behaviour Change Intervention	Behaviour Change Intervention Ontology (BCIO)
(Schenk et al., 2024)	Human Behaviours	Human Behaviour Ontology (HBO)

Existing modeling approaches can generally be divided into two categories. One category is knowledge-driven approaches, which mainly rely on ontologies and expert knowledge to explicitly model behavior ([Mojarad et al., 2020b](#); [Kim et al., 2019](#); [Taji and Lange, 2016](#); [Norris et al., 2017](#); [Schenk et al., 2024](#)). The other category is data-driven approaches, which mainly rely on empirical data to identify behavioral patterns through methods such as clustering and classification ([Khanfar et al., 2023](#); [Sun et al., 2017](#); [Millonig and Gartner, 2008](#)).

From the perspective of application domains, most studies focus on behavior modeling within a single domain, such as physical activity ([Kim et al., 2019](#)), driving behavior ([Khanfar et al., 2023](#); [Sun et al., 2017](#)), eating processes ([Taji and Lange, 2016](#)), or spatio-temporal behaviour ([Millonig and Gartner, 2008](#)), showing clear domain-specific characteristics. In contrast, a small number of studies have started to explore more general frameworks for representing human habits ([Norris et al., 2017](#); [Schenk et al., 2024](#)). Among these, the HBO is particularly noteworthy, as it aims to provide a unified ontological representation for more general human

behavior and therefore shows greater potential for cross-domain reuse and general behavior habit modeling ([Schenk et al., 2024](#)).

3 Research Design

The DSR paradigm was adopted because the objective of this research is not merely to design a simple prototype, but to synthesize problems in the field of AI-supported HRSs and to design, implement, demonstrate, and evaluate an artifact. In addition, to ensure a clear process and to provide a reference for the future design and evaluation of AI-supported HRSs, the meta-requirements, DPs, and DDs of this field were systematically identified and derived.

The artifact developed in this thesis is a RAG-enhanced, ontology-informed Habit Recommender System. It supports users in donating habits, structuring contextual information through an ontology, completing user forms, and constructing and modifying a user-defined knowledge base. Finally, based on user habits, the content provided in user forms, knowledge base retrieval results, and user goals, the system comprehensively generates personalized health-related recommendations.

The research process was organized according to the DSRM proposed by [Peffer et al. \(2007\)](#), as shown in Figure 1. DSRM connects problem identification, the definition of solution objectives, artifact design and development, demonstration, evaluation, and communication into a coherent research process ([Peffer et al., 2007](#)). Accordingly, the following sections are structured according to the main DSRM activities: Identify Problem & Motivate, Define Objectives of a Solution, Design & Development, Demonstration, and Evaluation. The final DSRM activity, communication, is realized through the written presentation of the research process, artifact design, implementation, demonstration, and evaluation results in this thesis, and is therefore not presented as a separate section ([Peffer et al., 2007](#)). In addition, the overall logical chain from meta-requirements to DPs, DDs, and the artifact is illustrated in Figure 2.

In the Identify Problem & Motivate stage, the current state of research on AI-supported HRSs was analyzed based on the Basics and Related Research section, including a non-systematic review of AI-supported HRS literature, a previous systematic literature review on habit modeling methods conducted in the research seminar, and additional relevant literature in the field of AI-supported HRSs. On this basis, and in combination with relevant review studies, the problems were identified ([Peffer et al., 2007](#)). Figure 3 provides a summarized overview of the problems currently faced by AI-supported HRSs, as well as the specific research problem further derived from these problems.

In the Define Objectives of a Solution stage, the identified problems were translated into meta-requirements. The solution objectives and the meta-requirements were treated as equivalent expressions at the same level of abstraction. The resulting meta-requirements define what an AI-supported HRS should achieve in order to address the identified problem space. At the same time, they serve as the basis for the evaluation goals used to evaluate the Habit Recommender

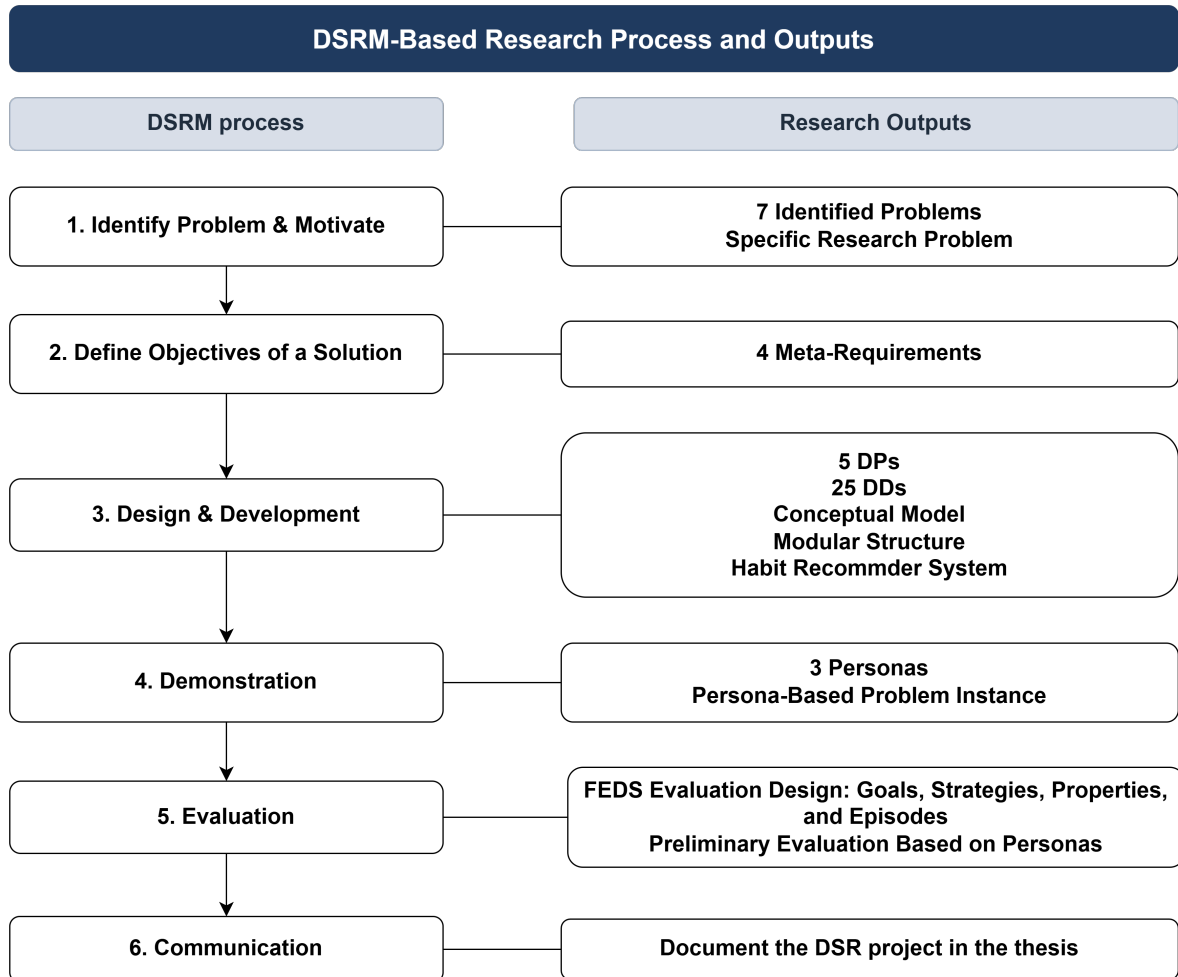
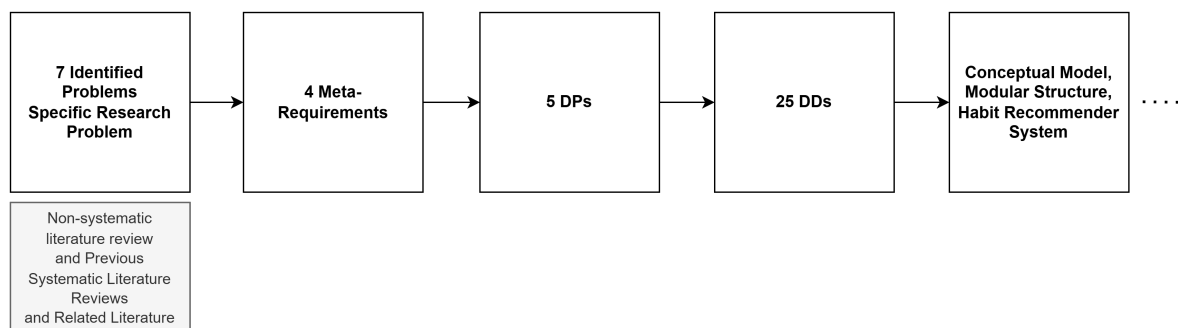


Figure 1: DSRM-Based Research Process and Outputs

Figure 2: Traceability Mapping from Literature-Informed Problem Identification to Meta-Requirements, DPs, DDs, and the Artifact (based on [Peffers et al. \(2007\)](#); [Kopenhagen et al. \(2012\)](#))

System prototype in the evaluation stage (Peffers et al., 2007). RQ1 is answered in this stage. As shown in Figure 4, the identified problems were translated into meta-requirements.

In the Design & Development stage, the meta-requirements were gradually translated into DPs and DDs and further transformed into the concrete artifact. First, DPs were derived from the meta-requirements to explain the general design logic that the artifact should follow (Peffers et al., 2007; Venable et al., 2016; Möller et al., 2020). Figure 5 shows this mapping. This step corresponds to RQ2.

Second, the DPs were further operationalized into traceable DDs to explain how these principles were instantiated in concrete system functions, architectural elements, data structures, and user interface components (Peffers et al., 2007). Figures 6, 9, 10, 11, and 12 show the mapping from DPs to DDs. This step corresponds to RQ3.

These DDs were implemented in the Habit Recommender System prototype. Based on this, the corresponding conceptual model was developed. The conceptual model can be understood as a high-level abstract representation of the key concepts in the system and their relationships, without addressing concrete implementation details (Tanriöver and Bilgen, 2011; Robinson et al., 2015). Figure 13 shows the conceptual model.

Subsequently, a modular design was implemented. The modules have clearly defined system roles and relatively clear input and output boundaries, and they support the implementation of the overall workflow. These modules also represent the concrete realization of the DDs (Parnas, 1972; Bass et al., 2022; IEEE, 2009). Figures 14, 15, and 16 show these modules, while Table 5 presents the mapping relationship between the modules and the DDs.

Finally, a prototype of the Habit Recommender System was implemented. The detailed implementation code is provided in the GitHub repository (Hua, 2026). Figures 17, 18, and 19 show the frontend pages.

In the demonstration stage, the artifact was applied to a concrete problem instance (Peffers et al., 2007). Three representative personas were created through a literature-informed persona construction approach and were used for the demonstration and evaluation (Tariq et al., 2012; Veldsman and Coetzee, 2022). Figure 20 shows the mapping relationship between the personas and the meta-requirements. A detailed overview of the three personas and the literature used for their construction is provided in Table 7.

To avoid repetition, Persona 1 was used as an example to demonstrate the complete process of entering user habits, completing and modifying forms, entering a goal, and generating recommendations. Figure 21 shows the recommendation result.

In the evaluation stage, the artifact was evaluated based on the previously defined meta-requirements. The evaluation design was organized with reference to the FEDS framework

proposed by [Venable et al. \(2016\)](#), resulting in evaluation goals, an evaluation strategy, evaluation properties, and evaluation episodes for each persona, as shown in Figure 22.

The detailed evaluation episodes for each persona are documented in the `EvaluationDesign` worksheet of the `Evaluation.xlsx` file. The concrete evaluation results are provided in the `EvaluationAllResults` worksheet of the same file in the GitHub repository ([Hua, 2026](#)).

Considering the limitations of time, resources, and the current stage of prototype development, only those properties of the meta-requirements that could be practically evaluated were assessed in a preliminary manner. Therefore, this research uses persona-based evaluation episodes to examine whether the artifact can meet the defined evaluation goals and properties. The results of the preliminary evaluation of the Habit Recommender System are presented through three concept matrices, namely Table 8, Table 9, and Table 10, thereby answering RQ4.

4 Habit Recommender System

4.1 Identify Problem & Motivate

According to the DSRM proposed by [Peppers et al. \(2007\)](#), the Identify Problem & Motivate stage first requires a clear definition of the specific research problem and a justification of the value of solving it. Based on the current state of AI-supported HRS research summarized in the Basics and Related Research section, the following discussion further synthesizes the core problems currently faced by this field in light of the relevant literature.

Problem 1 is narrow domain coverage, as existing HRSs provide only limited support beyond a few isolated health domains.

In terms of application domains, existing HRSs still remain largely concentrated in a few relatively independent and narrowly defined areas. Although such single-domain system designs may help reduce system complexity and allow systems to focus on specific health-related problems, relevant reviews indicate that the overall domain coverage of current HRSs remains limited, with research attention primarily centered on only a small number of therapeutic areas. Consequently, a system architecture capable of flexibly addressing a broader range of health scenarios has not yet been established ([Hors-Fraile et al., 2018](#); [Barbaric et al., 2025](#)).

Problem 2 is multidimensional mismatch, as existing HRSs fit poorly with the physical, cognitive, mental, and social complexity of health promotion.

However, single-domain HRSs are not well suited to the multidimensional nature of health promotion itself. [Lopez-Barreiro et al. \(2024a\)](#) explicitly note that health promotion encompasses multiple dimensions, including physical, cognitive, mental, and social aspects, and emphasize the need to foster healthy habits across all areas of life. This suggests that real-world health behavior change does not center on a single isolated behavior, but typically involves multiple interrelated dimensions ([Barbaric et al., 2025](#)). Therefore, if a system remains limited to a single domain, its recommendation scope and support capabilities are unlikely to adequately reflect the complexity of actual health management scenarios ([Lopez-Barreiro et al., 2024a](#); [Barbaric et al., 2025](#)).

Furthermore, recommendations restricted to a single domain are less likely to sustain the level of engagement, usage, and retention required for long-term behavior change. A relevant review notes that, although digital health interventions have the potential to promote positive health-related behavior change, they often suffer in practice from high dropout rates, low usage rates, and poor retention rates. Accordingly, improving user engagement and user experience is crucial to the effectiveness of such systems ([Cheung et al., 2019](#)). If an HRS is limited to providing recommendations within a single domain, it is less able to continuously adapt to

users' evolving needs in real-life contexts, thereby further undermining engagement and user experience.

Consequently, future HRSs should not remain confined to narrow domains and isolated recommendation tasks, but should evolve toward more comprehensive and holistic systems (Lopez-Barreiro et al., 2024a; Barbaric et al., 2025).

Problem 3 is inconsistent habit/context modeling, as there is still no consistent framework for representing habits and contextual information.

Existing HRSs have increasingly relied on multi-source behavioral data to support personalized recommendations, yet a consistent modeling framework for representing heterogeneous habits and their contextual information is still lacking (Hors-Fraile et al., 2018; Cheung et al., 2019; Barbaric et al., 2025).

The lack of a consistent modeling framework for habits and their contextual information makes it difficult to compare, reproduce, and build upon findings across different studies (Hors-Fraile et al., 2018). At the same time, when the representations of behaviors, habits, and contexts remain opaque, the underlying recommendation logic becomes even harder to interpret and validate (Hors-Fraile et al., 2018; Barbaric et al., 2025). This further reduces the interpretability of HRSs and may negatively affect user experience (Hors-Fraile et al., 2018). It may also hinder the effective integration of heterogeneous health-related data (Hors-Fraile et al., 2018; Cheung et al., 2019; Lopez-Barreiro et al., 2024a; Barbaric et al., 2025; Wang et al., 2024).

Problem 4 is weak NCD integration, as NCD-related information is often not embedded in recommendation logic.

NCD-related HRSs still remain relatively underexplored overall, and existing studies are concentrated on only a small number of disease topics (Barbaric et al., 2025). This is partly related to the fact that earlier health recommender research mostly focused on generic health promotion, meaning that many systems were originally designed primarily for general health promotion and behavior change rather than for constraint-based recommendation centered on specific NCDs (Cheung et al., 2019). However, with the expansion of globalization, rapid urbanization, and the widespread adoption of sedentary lifestyles, NCDs have become a global health problem and have imposed substantial social and economic burdens on both developed and developing countries (Abegunde et al., 2007). Against this background, if HRSs continue to treat NCDs merely as broad health goals or motivations for behavior change, without genuinely incorporating NCD-related information into the recommendation logic, they will struggle to adequately support disease management (Barbaric et al., 2025).

Problem 5 is incomplete feedback mechanisms, as adaptive feedback loops and profile updating remain limited.

Many HRSs are still in the early stages or at the prototype level, so feedback mechanisms have not yet been fully implemented ([Hors-Fraile et al., 2018](#)). Many systems essentially provide personalized content to users through message delivery, rather than building a continuous, interactive, and dynamically adaptive recommendation loop ([Cheung et al., 2019](#)). However, for HRSs to remain viable in the long term, they must evolve with their users; otherwise, the system will continue to rely on the user's initial state ([Hors-Fraile et al., 2018](#)). Implicit feedback expresses not only a user's potential preferences but also the system's confidence in its assessment of those preferences. If the system fails to make effective use of these feedback signals, it may distort the user's true profile ([Hu et al., 2008](#)).

Problem 6 is low interpretability, resulting in reduced transparency, trust, validation, and system improvement.

Many HRSs still lack sufficient interpretability for both users and researchers, which can have a range of adverse effects. From the user's perspective, low interpretability means that the system struggles to clearly present the reasoning and data basis underlying its recommendations, thereby reducing transparency and making it harder for users to determine when to trust, accept, or question a recommendation ([Herlocker et al., 2000](#); [Tintarev and Masthoff, 2012](#)). This not only undermines users' trust in and acceptance of the system but may also impair their ability to make high-quality decisions based on its recommendations ([Tintarev and Masthoff, 2012](#)). From the perspective of researchers and system designers, low interpretability also undermines the system's verifiability and improvability, making it more difficult to analyze the mechanisms underlying the recommendation logic, identify sources of error, and determine why the system is effective, for whom it is effective, and under what conditions it is effective ([Tintarev and Masthoff, 2012](#)).

Problem 7 is LLM risks and data limitations, including hallucination, sparsity, and cold-start problems.

Although LLMs bring stronger semantic understanding, generative capabilities, and interactive potential to recommendation systems, their use also introduces a significant risk: hallucination ([Rawte et al., 2023](#); [Zhao et al., 2024](#)). In recommendation scenarios, this problem may manifest in the generation of non-existent recommended items, the provision of descriptions that do not align with factual information, or the delivery of recommendations that appear plausible but are factually incorrect ([Zhao et al., 2024](#)). More importantly, hallucinations may have particularly severe consequences in high-stakes recommendation settings. For example, in contexts such as medical recommendations or legal advice, the dissemination of incorrect information may lead to serious real-world consequences ([Zhao et al., 2024](#); [Rawte et al., 2023](#)).

Existing HRSs still face sparsity and cold-start problems. These issues are closely related to

the fact that current recommendation systems generally rely on user–item historical interactions for personalized modeling: when users have limited historical records, the system finds it difficult to learn stable preference representations; and when new users or new items lack sufficient interactions, the system also finds it difficult to generate reliable recommendations ([Zhao et al., 2024](#)).

Based on the problems identified above, the specific research problem addressed is that current AI-supported HRSs still lack a unified and trustworthy design foundation, making it difficult to systematically integrate heterogeneous, cross-domain behaviors, habits, contextual information, NCD-related information, and user feedback, and to build interpretable personalized recommendations on that basis.



Figure 3: Identified problems and specific research problem in AI-supported HRSs

Figure 3 provides an overview of the core problems currently faced by AI-supported HRSs, as well as the specific research problem that is further derived from these problems.

4.2 Define Objectives of a Solution

In the “Define Objectives of a Solution” stage, the objectives of the solution are derived from the identified problems. In this thesis, the solution objectives are expressed in the form of meta-requirements. In other words, in the context of this thesis, Objectives of a Solution and meta-requirements represent different expressions of the same level of content (Peffer et al.,

2007; [Kopenhagen et al., 2012](#)).

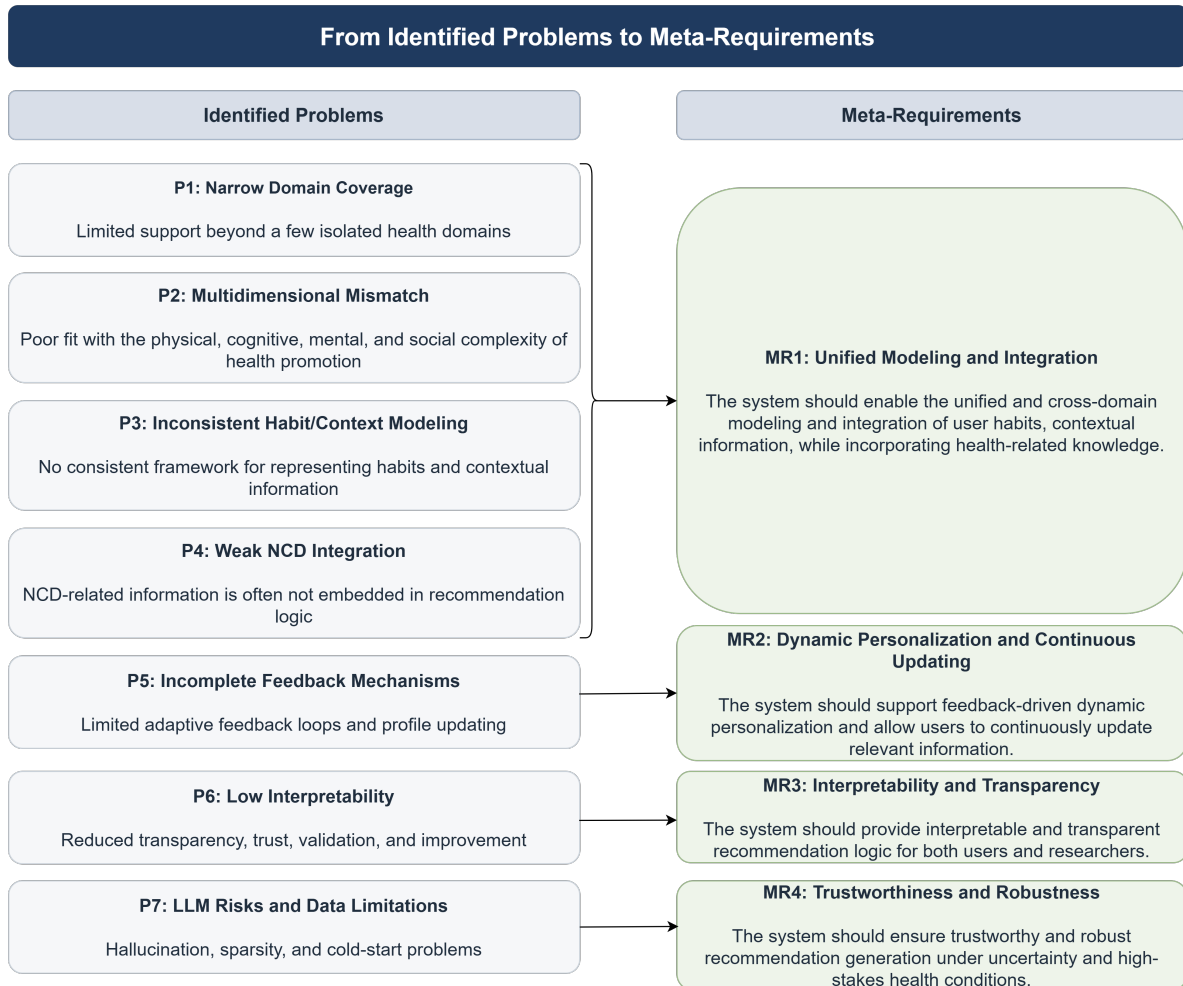


Figure 4: From Identified Problems to Meta-Requirements

First, P1 to P4 can be understood as different manifestations of a shared underlying need. Specifically, P1 reflects the limitation of the system in terms of its scope of application, namely that its support is usually restricted to a small number of isolated health domains. P2 reflects the inadequacy of the system in representing user profiles, that is, the system has difficulty adequately capturing the complex interactions among users' physical, cognitive, psychological, and social factors in the process of health promotion. P3 further reveals an important structural reason behind this problem, namely that current research lacks a consistent framework for representing habits and their contextual information, making it difficult to model user profiles in a sufficiently comprehensive and fine-grained manner. At the same time, P4 shows that this lack of modeling and integration capability further limits the ability to embed NCD-related

knowledge into the recommendation logic. It can therefore be seen that P1 to P4 do not point to several independent problems, but rather to one unified meta-requirement:

MR1: The system should enable the unified and cross-domain modeling and integration of user habits, contextual information, while incorporating health-related knowledge.

This meta-requirement implies that the solution should no longer address health recommendation problems in an isolated, partial, and domain-specific manner, but should instead provide a unified representational foundation for different domains, different health goals, and different types of user states.

P5 reveals that existing HRSs lack recommendation mechanisms capable of continuous interaction and dynamic updating. As a result, such systems often rely primarily on the user's initial state and struggle to adapt continuously to changes in user preferences, health status, behavioral habits, and contextual conditions. Given that behavior change is inherently a long-term, dynamic, and iterative process, the absence of feedback-driven adaptation significantly weakens the system's long-term usability and overall user experience ([Hors-Fraile et al., 2018](#); [Cheung et al., 2019](#); [Hu et al., 2008](#)). Consequently, a second meta-requirement can be derived: MR2: The system should support feedback-driven dynamic personalization and allow users to continuously update relevant information.

This meta-requirement emphasizes that the solution should effectively leverage both explicit and implicit user feedback, while also enabling users to continuously update their preferences, health status, behavioral habits, and contextual information.

P6 highlights the limited interpretability of existing HRSs. For users, low interpretability reduces the transparency of recommendations, making it difficult to understand the underlying rationale and evidence behind them, which in turn undermines trust and acceptance. For researchers and system designers, low interpretability hinders the analysis of recommendation logic, the identification of error sources, and the validation and improvement of system performance ([Herlocker et al., 2000](#); [Tintarev and Masthoff, 2012](#); [Rudin, 2019](#)). Therefore, a third meta-requirement can be proposed: MR3: The system should provide interpretable and transparent recommendation logic for both users and researchers. This meta-requirement implies that AI-supported HRS artifacts should clearly explain to users why recommendations are generated, while also making the system's design logic sufficiently transparent for researchers to analyze, validate, and improve it.

Finally, P7 reflects the risks of existing systems in terms of trustworthiness and robustness. On the one hand, although the integration of LLMs enhances semantic understanding, generation capabilities, and interactive potential, it also introduces the risk of hallucinations. On the other hand, existing HRSs still commonly suffer from sparsity and cold-start problems. When user

histories are insufficient, or when new users or new items lack interaction data, the system struggles to form stable preference representations and generate reliable recommendations (Rawte et al., 2023; Zhao et al., 2024; Wu et al., 2024; Ricci et al., 2011). Based on this, a fourth meta-requirement can be derived:

MR4: The system should ensure trustworthy and robust recommendation generation under uncertainty and high-stakes health conditions.

This meta-requirement emphasizes that the solution should minimize, as far as possible, the risks arising from hallucinations, data sparsity, cold start, and other sources of uncertainty, so that the recommendations generated by the system remain sufficiently trustworthy and robust in high-stakes health contexts.

As shown in Figure 4, the seven identified problem categories are synthesized into four higher-level meta-requirements, namely MR1 (Unified Modeling and Integration), MR2 (Dynamic Personalization and Continuous Updating), MR3 (Interpretability and Transparency), and MR4 (Trustworthiness and Robustness), which provide the core foundation for the subsequent derivation of DPs and DDs (Peffer et al., 2007; Koppenhagen et al., 2012).

4.3 Design & Development

In the design and development stage, the meta-requirements defined in the previous stage are further transformed into a concrete artifact. According to Peffer et al. (2007), the core of this stage lies in the creation of the artifact, that is, in determining its intended functionality, defining its architecture, and implementing its concrete instantiation. Based on this general understanding, the logic of design principle development proposed by Möller et al. (2020) suggests that meta-requirements, as high-level goals and abstract requirements for a class of artifacts, can be further translated into DPs. These DPs are not simple repetitions of the meta-requirements, but rather principled design responses to them. In other words, DPs are not mere restatements of the MRs; instead, they represent mid-level design knowledge that answers the question of what principles the system should follow in its design in order to satisfy these meta-requirements. Subsequently, according to Koppenhagen et al. (2012), these abstract DPs need to be further operationalized into specific DDs and ultimately embodied in corresponding artifact elements, so that the transformation process from meta-requirements to DPs, and then to DDs and the artifact, remains explicit and traceable.

Therefore, the focus of this stage is to derive DPs from the meta-requirements proposed in the previous subsection, then to derive traceable DDs from these DPs, and on this basis to design the conceptual model and modular structure of the prototype, which is finally implemented as a concrete system prototype.

4.3.1 Derivation of DPs from Meta-Requirements

As shown in Figure 5, four meta-requirements were identified, and five DPs were derived based on them. Among them, MR1, “Unified Modeling and Integration,” was further developed into two interrelated DPs. This is because Unified Modeling and Integration involves, on the one hand, the unified semantic representation of heterogeneous habit data and contextual information and, on the other hand, the integration of multidimensional user information at the level of user modeling. Therefore, two separate DPs were required. DP1.1, “Semantic Unification,” emphasizes that the system should establish a shared semantic representation in its design in order to support the unified and cross-domain modeling of heterogeneous habit data and contextual information. DP1.2, “Context-Rich User Modeling,” emphasizes that the system should construct user profiles from multiple dimensions, including daily behaviors, mental states, health goals, and NCD-related information, so that recommendations can reflect the multidimensional nature of behavior change ([Barbaric et al., 2025](#); [Lopez-Barreiro et al., 2024a](#)). MR2, “Dynamic Personalization and Continuous Updating,” is further translated into DP2, “Feedback-Driven Adaptivity,” meaning that the system should be able to continuously adapt recommendation results and user profiles based on explicit feedback or implicit feedback, and user-initiated updates ([Rendle et al., 2012](#); [Hu et al., 2008](#); [Oard and Kim, 1998](#)). MR3, “Interpretability and Transparency,” is transformed into DP3, “Inspectable Recommendation Logic,” which emphasizes that, by design, the recommendation process should be made inspectable through traceable design logic, supporting evidence, and user-facing explanation mechanisms ([Tintarev and Masthoff, 2012](#); [Herlocker et al., 2000](#); [Rudin, 2019](#)). Finally, MR4, “Trustworthiness and Robustness,” is transformed into DP4, “Robust Recommendation Design,” which emphasizes that the system should incorporate mechanisms in its design to support the generation of trustworthy and robust recommendation results under uncertainty and high-stakes health conditions ([Wu et al., 2024](#)).

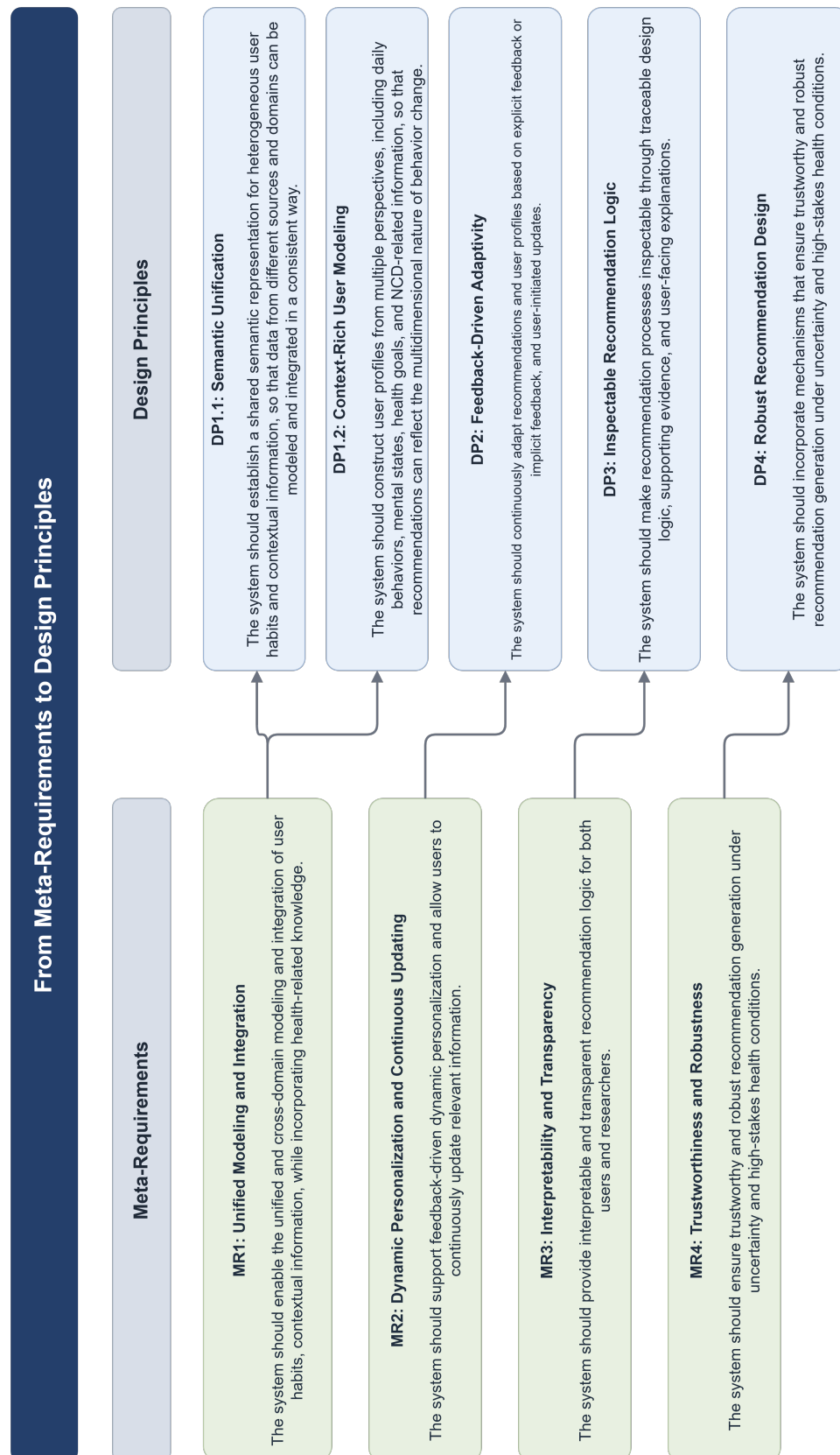


Figure 5: From Meta-Requirements to DPs

4.3.2 Derivation of Traceable DDs

DDs are not merely a simple refinement of DPs; rather, they should be explicitly specified as concrete instantiations of the DPs. The transformation from DPs to DDs typically exhibits a one-to-many relationship. Although a DD may simultaneously support multiple DPs, it should at least be explicitly assigned to one of them (Kopenhagen et al., 2012). At the same time, DDs can take the form of various concrete design elements, such as use cases, features, system functions, architectural concepts or elements, user interface patterns or elements, and integration patterns, thereby providing a direct basis for the subsequent implementation of the artifact (Peffer et al., 2007; Kopenhagen et al., 2012).

As shown in Figure 6, DP1.1 was further operationalized into five specific DDs.

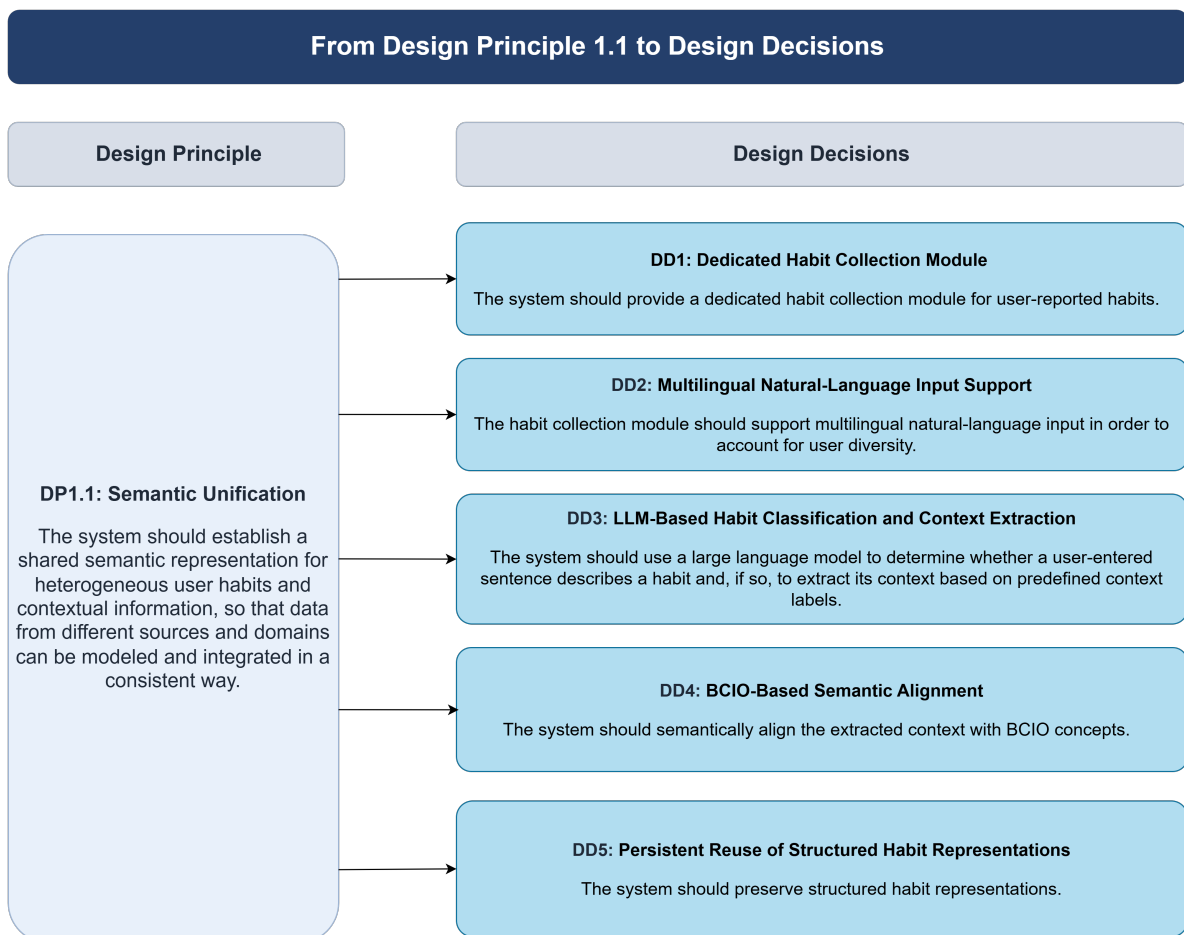


Figure 6: From Design Principle 1.1 to DDs

With regard to DD1, “Dedicated Habit Collection Module,” if the system aims to achieve the unified modeling of heterogeneous user habits and contextual information, it must first provide a

dedicated and stable data entry point. In this way, users can more clearly recognize that they are in the process of donating their habits.

DD2, “Multilingual Natural-Language Input Support,” takes into account that heterogeneous data are reflected not only in data sources, but also in forms of linguistic expression. Different users may use different languages, different wordings, and different levels of expressive granularity. If the system supports only a single language, then although the degree of structuring may appear to be improved on the surface, the authenticity of user expression would in fact be sacrificed. LLMs possess such multilingual capability, because they are pretrained on highly diverse corpora (Zhu et al., 2024).

Once the system receives user input in natural-language form, merely having the raw text is not sufficient to achieve semantic unification. The system must also determine whether the input actually describes a habit. If so, it must identify which contextual elements it contains that can be further modeled. This is also why DD3, “LLM-Based Habit Classification and Context Extraction,” is needed.

The contextual information of habits is highly important for behavior change. Changing the context of an existing bad habit can help disrupt that habit, whereas repeatedly performing a behavior in a stable context can help establish stable context–response associations and thereby support the formation of a good habit (Stojanovic et al., 2022; Wood and Neal, 2007). The reason for choosing LLMs here is that, in the author’s course paper (Belegarbeit), multiple LLMs were tested to examine whether they could accomplish the tasks of habit classification and context classification, across English and German and under different prompt strategies. The LLMs used included the Claude series (Anthropic, 2025), the DeepSeek series (DeepSeek-AI et al., 2024, 2025a,b), OpenAI models (OpenAI, 2025), and others.

The results showed that, on a small dataset of 66 sentences, as illustrated in Figure 7, most LLMs were able to correctly classify habits in both English and German under zero-shot conditions, that is, without any examples.

As shown in Figure 8, most models also performed well on the context classification task in the zero-shot setting. Therefore, LLMs were also retained for LLM-based habit classification and context extraction. Traditional supervised models require large amounts of high-quality labeled data and are limited in their ability to extract complex contextual information. In contrast, LLMs are better suited to handling open-ended text, implicit semantics, and generalization across different forms of expression (Pai et al., 2024; Plank and Moschitti, 2013).

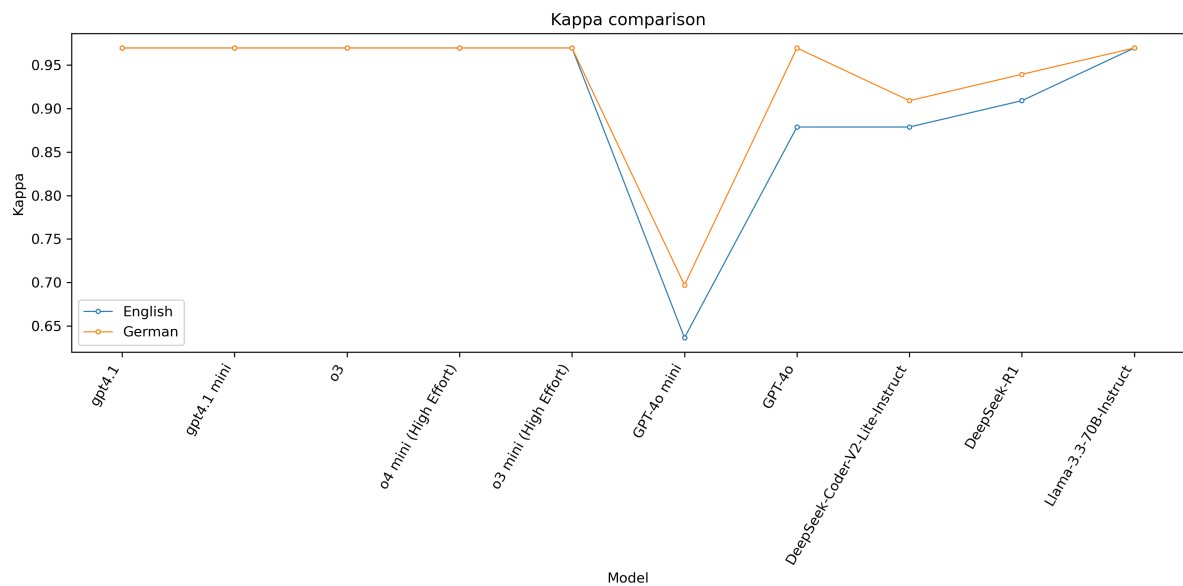


Figure 7: LLM performance (Kappa) on the habit classification task in a zero-shot setting for the English and German datasets

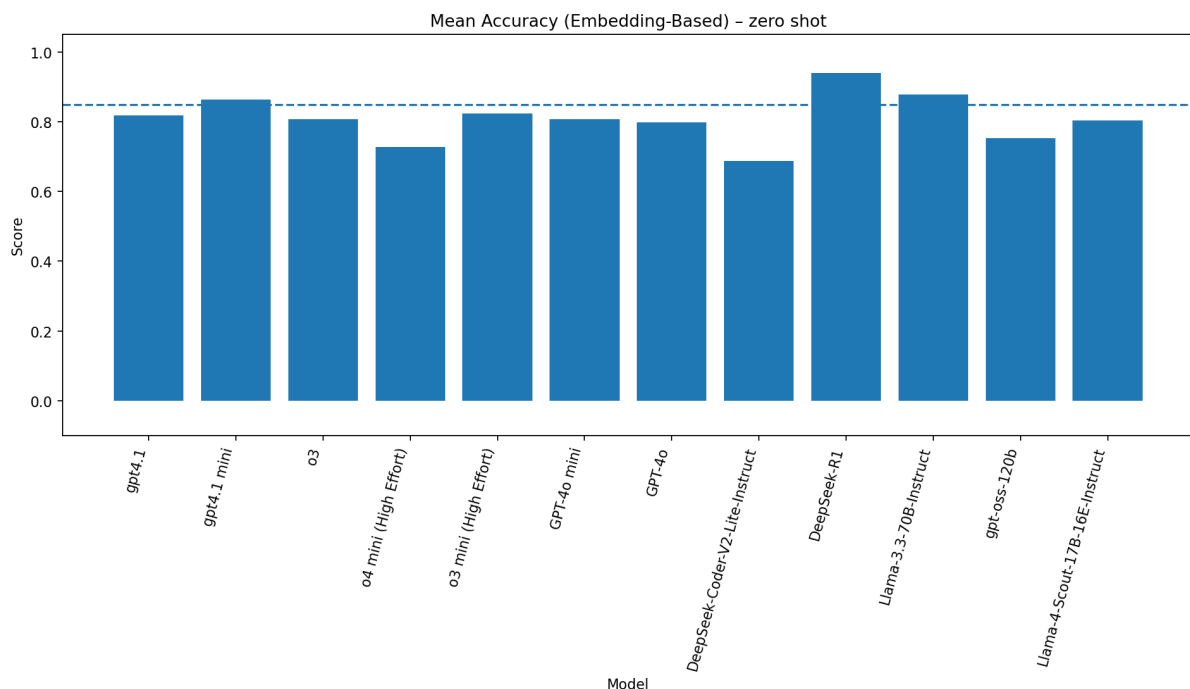


Figure 8: Mean accuracy (embedding-based) of different models in the zero-shot context extraction task

DD4, “BCIO-Based Semantic Alignment,” means that habits and contextual information in the system should not remain only at the level of surface words, but should be further aligned

with a unified, reusable, and interpretable conceptual system. The reason for choosing BCIO rather than other existing ontologies is, first, its thematic fit. As shown in Table 3, both in the previous research seminar and in the current non-systematic literature review on AI-supported HRSs, several candidate ontologies potentially relevant to habit modeling can be identified. However, from the perspective of thematic applicability, most of these ontologies focus on specific behavioral domains or application scenarios, such as human activity, physical activity, eating behaviour, food recommendation, or runner training (Mojarad et al., 2020a,b; Kim et al., 2019; Taji and Lange, 2016; Donciu et al., 2011; Hamdollahi Oskoue and Hashemzadeh, 2023). Such ontologies exhibit clear domain-specific characteristics and are therefore difficult to use directly as the unified semantic foundation for the cross-contextual and cross-domain habit recommendation system developed. In contrast, BCIO is not limited to one specific type of behavior, but instead models the higher-level conceptual framework of behaviour change intervention (Norris et al., 2017). Moreover, BCIO is the parent ontology of HBO, and HBO is intended to model all human behavior; thematically, it is therefore also closer to habits, which can themselves be understood as behaviors that have established context–response associations (Norris et al., 2017; Schenk et al., 2024; Wood and Neal, 2007). Since the aim is not merely to provide a static description of one type of behavior, but rather to connect user habits, contextual information, goals, and the subsequent recommendation process in the context of NCD prevention, BCIO is thematically more suitable.

Table 3: Candidate Ontologies Potentially Relevant for Habit Modeling Identified in the Previous Research Seminar and in the Current Non-systematic Literature Review on AI-supported HRSs

Reference	Modeled Object	Ontology Name
(Chatterjee et al., 2022a)	Daily Activity and Preferences	Semantic Ontology Model
(Chatterjee et al., 2022b, 2021)	Activity, Context, and Health State	UiA eHealth Ontology (UiAeHo)
(Mojarad et al., 2020a)	Human Activity and Context	Human Activity Ontology (HAVON)
(Donciu et al., 2011)	Runner Nutrition and Training	Nutrition Ontology + Sports Medicine Ontology + Runners Ontology

Table 3: Continued.

Reference	Modeled Object	Ontology Name
(Hamdollahi Oskouei and Hashemzadeh, 2023)	Food and User Profiles	Food and User Ontology
(Schenk et al., 2024)	Human Behaviour	HBO
(Mojarad et al., 2020b)	Human Activity	Human Activity Ontology (HAT)
(Kim et al., 2019)	Physical Activity	Physical Activity Ontology
(Taji and Lange, 2016)	Eating Behaviour	uc_Eating Ontology
(Norris et al., 2017)	Behaviour Change Intervention	BCIO

To further demonstrate the suitability of BCIO, Table 4 presents the following example: “When I wake up every morning and feel a bit tired, I go to the kitchen and drink a glass of water with the roommates who also happen to be awake, because I want to stay healthy and concentrate better during the day.” This example contains all contextual components of a habit. The extracted context labels include Time, Physical Setting, Prior Behavior, Other People, Internal State, and Goal/Reasoning. Based on the relevant literature, these contextual elements of habits were identified, and the corresponding phrases were extracted. These phrases can then be mapped to the corresponding classes or relations in BCIO.

Source text	label	value	HBO (Schenk et al., 2024)
When I wake up every morning and feel a bit tired, I go to the kitchen and drink a glass of water with the roommates who also happen to be awake, because I want to stay healthy and concentrate better during the day.	time (Wood and Rünger, 2016)	every morning	temporal region (BFO:0000008)
	physical settings / performance locations (Wood and Neal, 2007; Wood, 2017)	in the kitchen	location (BCIO:006085)
	preceding actions (Wood and Neal, 2007; Wood, 2017)	after waking up	individual human behaviour (BCIO:036000) with Precedes (BFO:0000063)
	particular people (Wood and Neal, 2007; Wood, 2017)	with the roommates who also happen to be awake	human population (BCIO:041000)
	internal states (Wood, 2017)	feeling tired	emotionally driven (BCIO:050814)
	goal / intention / reasoning (Wood and Rünger, 2016)	to stay healthy and concentrate better	goal (BCIO:006049)
	behavior (Wood and Rünger, 2016)	drink a glass of water	individual human behaviour (BCIO:036000)

Table 4: Example of mapping habit-related contextual elements to the BCIO

DD5, “Persistent Reuse of Structured Habit Representations,” means that semantic unification should not merely be a one-time processing result, but should instead become a foundational resource that can be continuously reused by the system. If structured habit representations were to be generated again from the original text each time they are needed, this would likely lead to problems such as repeated computation and inconsistent results (Atil et al., 2025; Regmi and Pun, 2024).

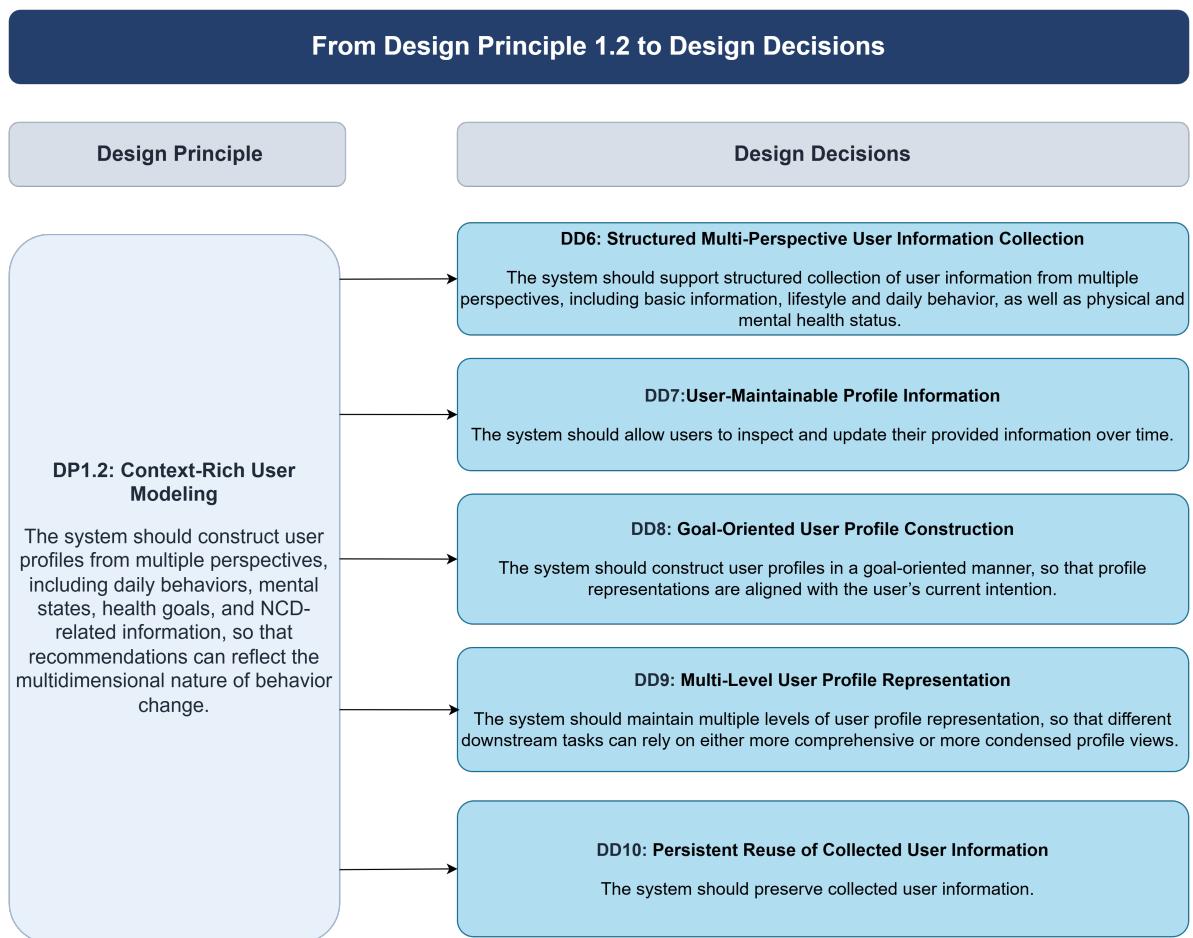


Figure 9: From Design Principle 1.2 to DDs

As shown in Figure 9, DP1.2 was further operationalized into five specific DDs.

DD6, “Structured Multi-Perspective User Information Collection,” emphasizes that user profile information should be collected from multiple perspectives, including basic information, lifestyle and daily behavior, as well as physical and mental health status (Chatterjee et al., 2022a,b). Collecting only a single type of information, for example only habits or only basic demographic information, would be a simpler approach with lower implementation costs. However, the problem is that this would result in only a very thin user profile and would be unable to reflect the interactions among different factors in behavior change, that is, the complexity of behavior change (Barbaric et al., 2025).

DD7, “User-Maintainable Profile Information,” is based on the fact that behavior change is itself a dynamic process. Users’ health goals, lifestyle habits, mental states, and even external living conditions may change over time (Zhang et al., 2025). If the user profile cannot be maintained, then user information would only be collected once at the initial stage and

subsequently assumed to remain unchanged, which would make it difficult to reflect the user's actual state. In addition, an editable user profile enables users to influence recommendation results by modifying their own information and thereby changing the profile, while also helping to improve recommendation quality (Mysore et al., 2023).

DD8, "Goal-Oriented User Profile Construction," emphasizes that the user profile should be constructed in a goal-oriented manner so that the profile representation can reflect the user's goals. This is important because goals play a significant role in behaviour change, and habits and goals interact in the process of shaping behaviour (Wood and R nger, 2016; Carden and Wood, 2018). In addition, research in the recommender systems field also highlights the value of goal- and intent-aware recommendation design, especially in situations where recommendations need to respond promptly to users' current needs (Habets and Coninx, 2025; Jannach and Zanker, 2025).

DD9, "Multi-Level User Profile Representation," refers to the use of different forms and levels of user profile representation for different downstream tasks. For example, more detailed and human-readable user profiles can be used for the presentation or explanation of recommendation results. Natural-language user profiles can enhance the transparency of recommender systems and provide users with the possibility to inspect recommendation results (Radlinski et al., 2022). By contrast, in RAG retrieval, the original user input is not always the optimal retrieval query, and complex queries may introduce noise. Therefore, it is necessary to further transform the detailed user profile into a more concise, retrieval-oriented profile representation (Ma et al., 2023). Therefore, maintaining multi-level user profile representations helps balance transparency and retrieval effectiveness.

DD10, "Persistent Reuse of Collected User Information," means that the system should retain collected user information in order to support its continuous reuse in subsequent stages. This is not only important for enabling later modules to reuse the information, but also for preserving a basis to analyze how user profiles evolve over time.

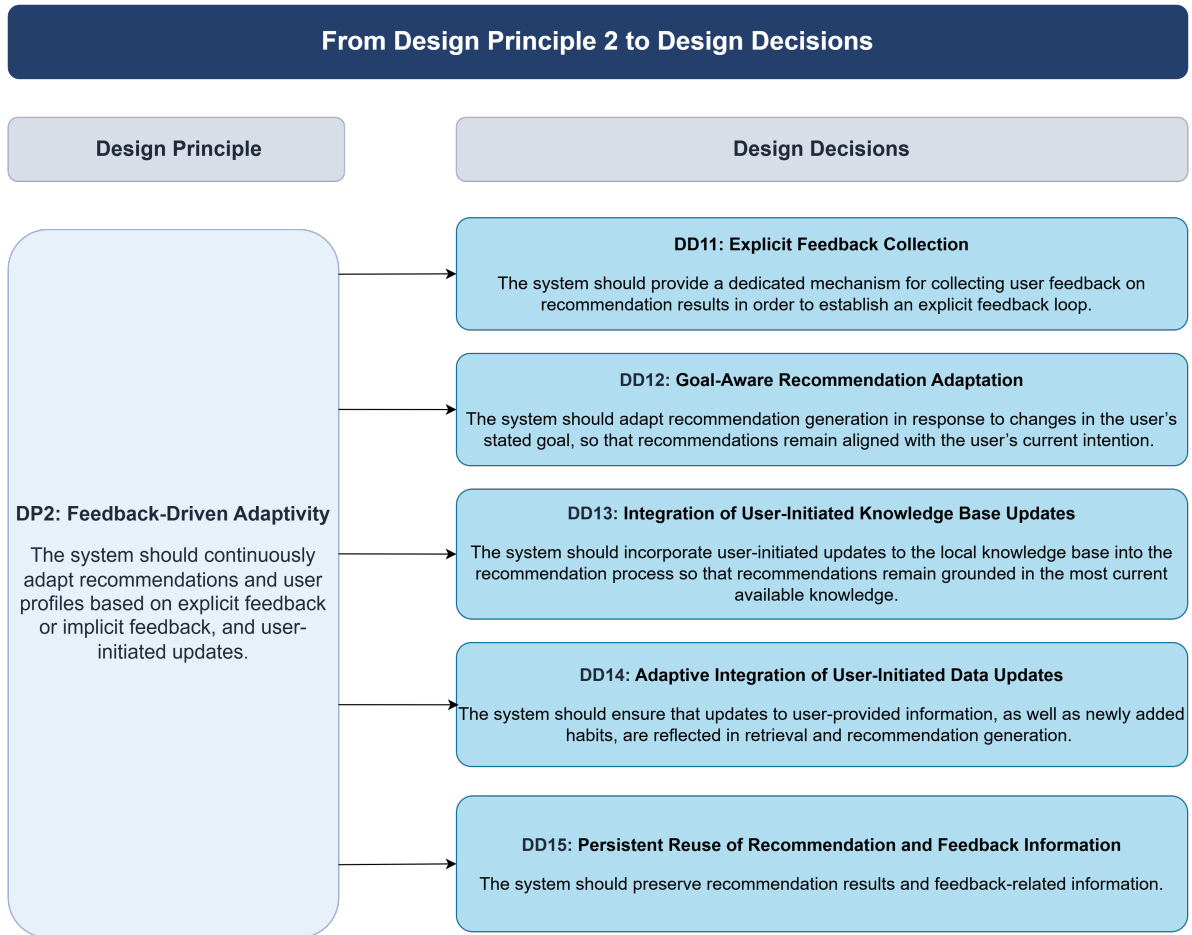


Figure 10: From Design Principle 2 to DDs

As shown in Figure 10, DP2 was further operationalized into five specific DDs.

DD11, “Explicit Feedback Collection,” is the most direct instantiation of the keyword “feedback-driven” in DP2. In recommender systems research, clicks, dwell time, and behavioral logs are commonly regarded as typical forms of implicit feedback (Hu et al., 2008; Rendle et al., 2012; Oard and Kim, 1998). However, in the relatively high-risk and highly explanation-sensitive context of health habit recommendation, users’ explicit feedback, such as whether a recommendation is “useful or not useful,” “appropriate or inappropriate,” or whether they are “willing or unwilling to try it,” helps provide the system with a more direct, more interpretable, and more easily traceable basis for adjustment, while also enhancing user control over the recommendation process (Ge et al., 2025). At the same time, an explicit feedback mechanism is also more consistent with the needs for system transparency and explainability emphasized (Tintarev and Masthoff, 2012).

DD12 reflects the requirement of “adapt recommendations” in DP2, but further clarifies what

the system should adapt recommendations around. The reason why a recommender system needs to be intent-aware is that, when users access the system at different points in time, their current usage intentions may change. Therefore, the system should consider, as far as possible, the user's current intention at a specific moment, rather than relying only on relatively stable long-term preferences when generating recommendations ([Jannach and Zanker, 2025](#)). Compared with many latent states that are more difficult to observe consistently, goals are high-value signals that users can explicitly express and that can be directly used to guide the direction of recommendations. Especially in the context of health behavior change, goal-oriented recommendation design is considered capable of improving the acceptance and persuasiveness of recommendations, because recommendations should remain aligned with the patient's current behavior, short-term goals, and long-term health goals ([Habets and Coninx, 2025](#)). Therefore, even if other profile information remains unchanged, the system should generate different recommendation results as soon as the user's current goal changes. Otherwise, the recommendations may become misaligned with the user's current purpose, thereby weakening their relevance and acceptability ([Habets and Coninx, 2025](#); [Jannach and Zanker, 2025](#)).

DD13, "Integration of User-Initiated Knowledge Base Updates," extends the "adaptivity" of DP2 from the user level to the knowledge level. If the recommender system relies on a local knowledge base to generate suggestions, then adaptivity should not only be reflected in the idea that "when the user changes, the recommendations change," but also in the idea that "when the knowledge base changes, the recommendations should change accordingly." In addition, users themselves should be able to create, read, update, and delete knowledge base content so as to build a personalized knowledge base that fits their own needs. The reason for choosing "integration of user-initiated knowledge base updates," rather than "continuous automatic acquisition of new knowledge from external networks by the system," is the emphasis on controlled, traceable, and verifiable knowledge updates, thereby ensuring that recommendations are always grounded in a current and trustworthy knowledge base ([Gao et al., 2024b](#); [Lála et al., 2023](#); [Lewis et al., 2021](#)).

DD14, "Adaptive Integration of User-Initiated Data Updates," focuses on the idea that user-initiated updates to personal information, as well as newly added habits, should be incorporated into the retrieval and recommendation process in a timely manner. DD12 focuses on changes in goals, DD13 focuses on changes in the knowledge base, and DD14 focuses on changes in the user's own data and habit-related information. Although all three belong to the broader category of "updates," the objects being updated are different, and therefore they need to be modeled separately. The reason for choosing "integration of user-initiated data updates," rather than "periodic profile reconstruction by the system" or "pure sensor-based automatic updating," lies

in the fact that the application scenario of the present study is primarily based on users donating information about their habits.

DD15, “Persistent Reuse of Recommendation and Feedback Information,” provides temporal continuity for DP2. If each round of recommendation were generated solely on the basis of the current input, without retaining previous recommendation results and user feedback, the system might still be able to produce recommendations for a single interaction, but it would be unable to improve recommendation generation over time through the reuse of accumulated information across multiple interaction rounds.

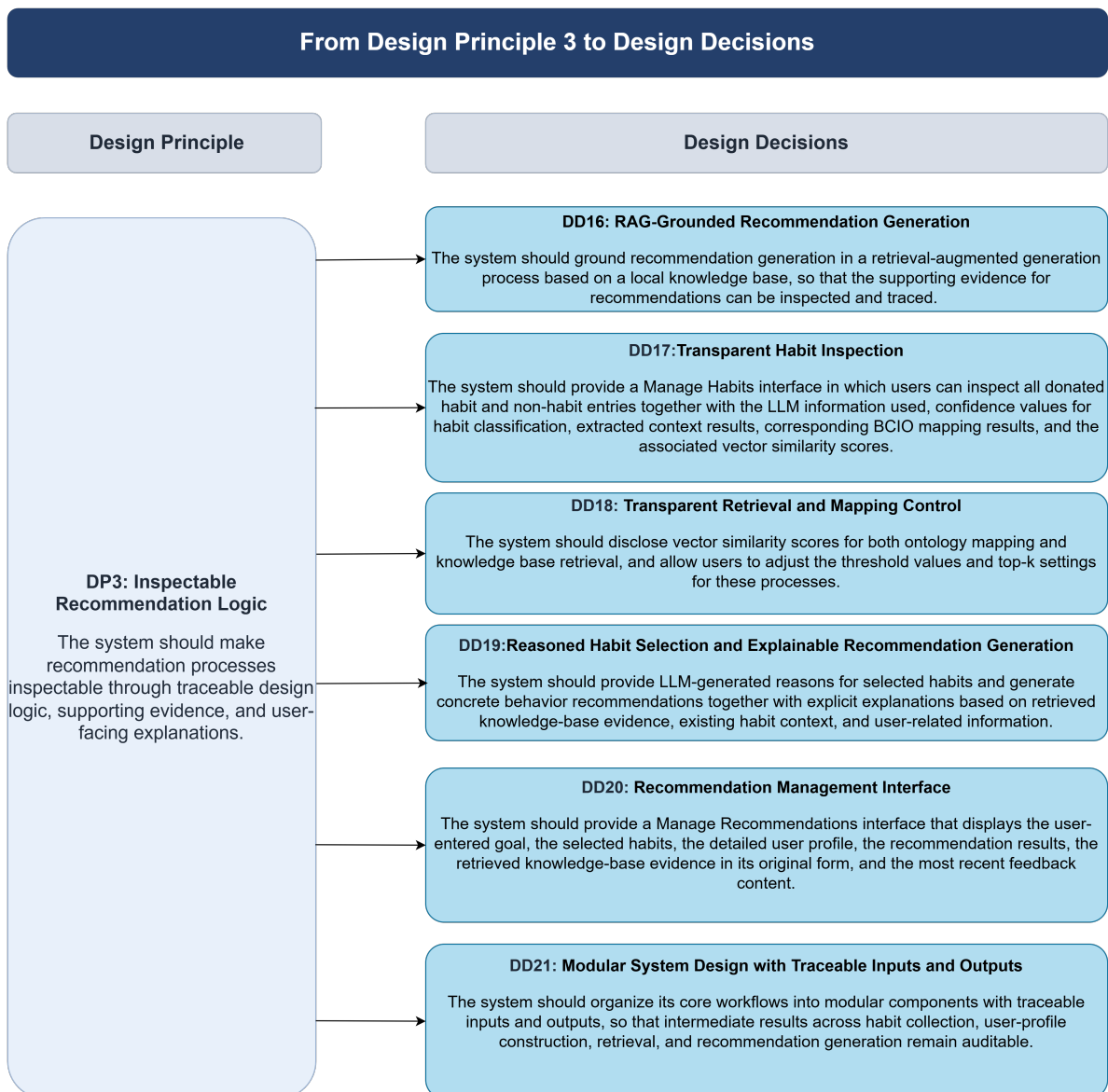


Figure 11: From Design Principle 3 to DDs

As shown in Figure 11, DP3 was further operationalized into six specific DDs.

DD16, “RAG-Grounded Recommendation Generation,” is a direct response to the requirement of “supporting evidence” in DP3. If recommendation results are generated directly by a black-box model, both users and researchers can hardly determine on what knowledge the recommendations are actually based (Rudin, 2019). Therefore, recommendation generation is grounded in a RAG pipeline based on a local knowledge base, so that recommendation results can be linked to externally retrievable and inspectable evidence. In this way, the design better satisfies DP3’s requirement for inspectability (Gao et al., 2024a; Liu, 2025).

DD17, “Transparent Habit Inspection,” further shifts the inspectability required by DP3 to the habit data level. Recommendation results are not generated out of nowhere, but are built on the basis of user-donated habits, habit classification results, context extraction results, and ontology mapping results. If the outputs of these preceding steps are themselves not visible, then even if the final recommendation is accompanied by explanations, the overall recommendation logic still remains incomplete. Therefore, the system provides a Manage Habits interface through which users can inspect donated habit and non-habit entries, LLM-based classification information, confidence scores, context extraction results, BCIO mapping results, and the corresponding vector similarity scores. This enhances the transparency of the system (Tintarev and Masthoff, 2007; Guesmi et al., 2022; Ramos et al., 2024).

DD18, “Transparent Retrieval and Mapping Control,” reflects an extension of the “traceable design logic” required by DP3 to the parameter level. If the system adopts vector retrieval and ontology mapping mechanisms, then one of the key decisions behind the recommendation lies in what was retrieved, why it was retrieved, and which candidates were retained or excluded due to threshold settings. To address this, the system not only discloses the similarity scores for ontology mapping and knowledge base retrieval, but also allows users to adjust key parameters such as threshold and top-k. The reason for choosing this design decision, rather than completely encapsulating these retrieval and mapping mechanisms as invisible internal processes, is that although the latter would be simpler to implement, it would weaken the inspectability of the recommendation logic (Luo et al., 2023; Cormack et al., 2009).

DD19, “Reasoned Habit Selection and Explainable Recommendation Generation,” is the core instantiation of the requirement for “user-facing explanations” in DP3. DP3 not only requires the internal system logic to be traceable, but also requires the system to explain to users why these habits were selected and why these recommendations were generated. Therefore, the system is required to generate explicit reasons for the selected habits and, during recommendation generation, to provide explicit explanations by combining retrieved knowledge base evidence, existing habit contexts, and relevant user information. The reason for choosing this design

decision, rather than merely displaying raw evidence snippets or providing only the final recommendation text, is that simply presenting evidence does not amount to an explanation, and merely giving the result provides even less insight into the underlying reasoning ([Tintarev and Masthoff, 2012](#); [Herlocker et al., 2000](#); [Rudin, 2019](#)). By contrast, habit selection with stated reasons and explainable recommendation generation can organically connect evidence and recommendation results, thereby responding more fully to the requirements of DP3.

DD20, “Recommendation Management Interface,” integrates the inspectability required by DP3 into a unified interaction interface. The system introduces a Manage Recommendations interface to centrally display the user’s input goal, the selected habits, the detailed user profile, the recommendation results, the original knowledge base evidence, and the most recent feedback. A unified recommendation management interface can organize what would otherwise be fragmented inputs, evidence, and outputs into a coherent and inspectable object, thereby providing stronger support for DP3.

DD21, “Modular System Design with Traceable Inputs and Outputs,” provides the structural foundation for implementing DP3. The reason why recommendation logic can be inspected depends not only on interface presentation, but also on whether the system itself is organized as an auditable modular process. The system decomposes the core process into modules with traceable inputs and outputs, so that the intermediate results of each stage can be retained and inspected. Only when module boundaries are clear and inputs and outputs are explicit does the inspectability required by DP3 become technically feasible ([Parnas, 1972](#)).

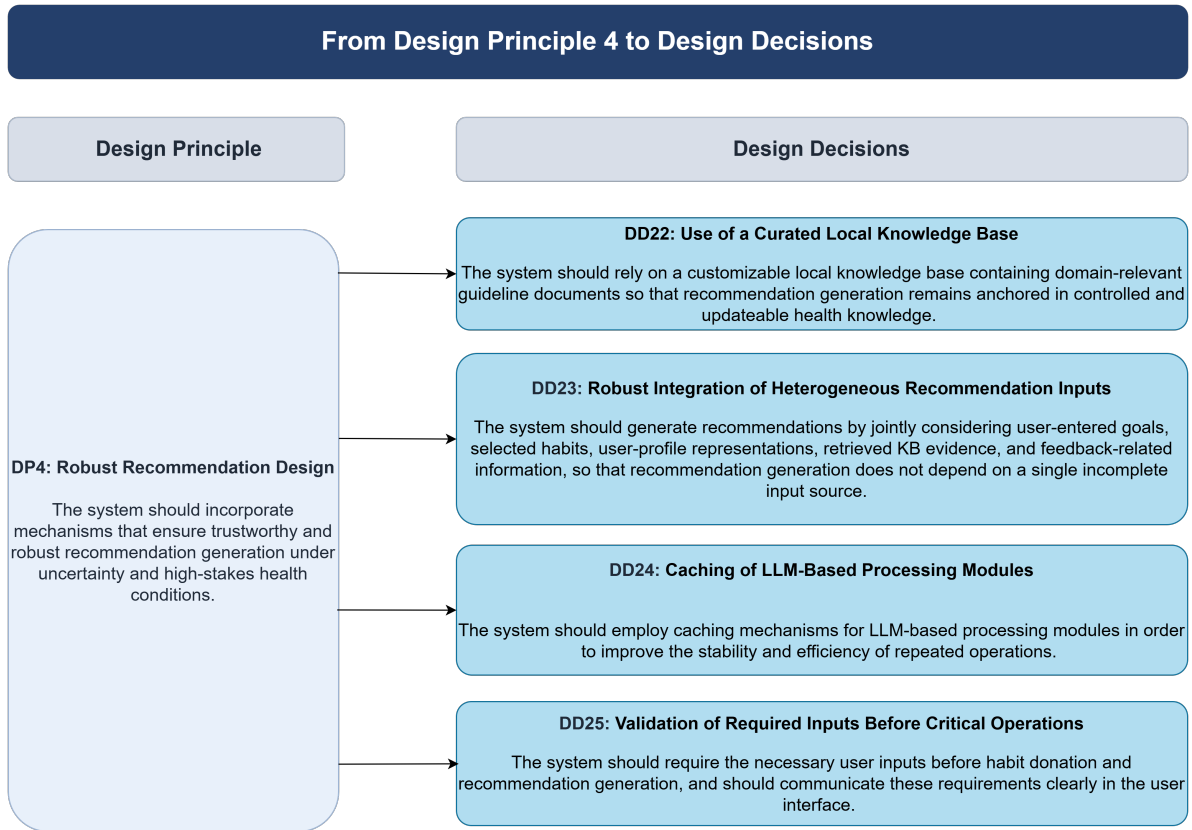


Figure 12: From Design Principle 4 to DDs

As shown in Figure 12, DP4 was further operationalized into four specific DDs.

DD22, “Use of a Curated Local Knowledge Base,” is a direct response to the requirements of trustworthiness and robustness in DP4. In the context of health recommendation, if the knowledge sources on which recommendation generation relies are not controlled, the system may be affected by varying information quality on the web, unclear sources, or content drift, thereby weakening the trustworthiness of the recommendation results (Wu et al., 2024; Sezgin and Özkan, 2013; Lopez-Barreiro et al., 2024b). By contrast, introducing a local knowledge base can effectively mitigate hallucinations in LLM-generated recommendations (Rawte et al., 2023; Lála et al., 2023).

DD23, “Robust Integration of Heterogeneous Recommendation Inputs,” reflects the requirement of “under uncertainty” in DP4. If recommendation generation relies excessively on a single input source, then once that input is incomplete, inaccurate, or biased, the final result becomes fragile (Dong et al., 2025). Therefore, the system is required to jointly consider the user’s input goal, the selected habits, the user profile representation, the retrieved knowledge base evidence, and feedback-related information, so as to avoid recommendation logic being dominated by a

single information source.

DD24, “Caching of LLM-Based Processing Modules,” extends the robustness required by DP4 to the system operation level. Since several core processes rely on LLM modules, such as habit classification, context extraction, or recommendation generation, repeatedly calling the model for every repeated operation would not only increase response time and resource consumption, but might also affect the consistency of repeated tasks due to fluctuations in model outputs (Regmi and Pun, 2024; Atil et al., 2025). Therefore, a caching mechanism is introduced for LLM-based processing modules in order to improve the stability and efficiency of repeated operations (Regmi and Pun, 2024).

DD25, “Validation of Required Inputs Before Critical Operations,” addresses the risk-control requirement of DP4 in high-stakes health contexts. Allowing habit donation or recommendation generation to proceed without critical inputs may lead to outputs that rest on incomplete or potentially misleading premises, thereby weakening the trustworthiness of the recommendations and increasing the likelihood of error (Tintarev and Masthoff, 2007; Guesmi et al., 2022; Ramos et al., 2024). To mitigate this risk, the system is required to validate required inputs before critical operations, to alert users when essential information is missing, and to communicate these requirements clearly in the user interface.

4.3.3 Conceptual Design

A Conceptual Model can be understood as a high-level abstract representation of the key concepts within a system and their interrelationships, which is used to describe the system’s main objects, structural relationships, and relevant behavioral characteristics without involving concrete implementation details (Tanrıöver and Bilgen, 2011; Robinson et al., 2015). In DSR, system design does not need to take the form of a fully instantiated artifact from the outset. Rather, DSR contributions can exist at different levels of artifact abstraction; since a model itself constitutes a type of artifact, a Conceptual Model can also be regarded as a legitimate form of artifact. Furthermore, even abstract meta-artifacts that have not yet been fully tested can still constitute valid DSR contributions (Peffer et al., 2007; Hevner and Chatterjee, 2010; Gleasure, 2014).

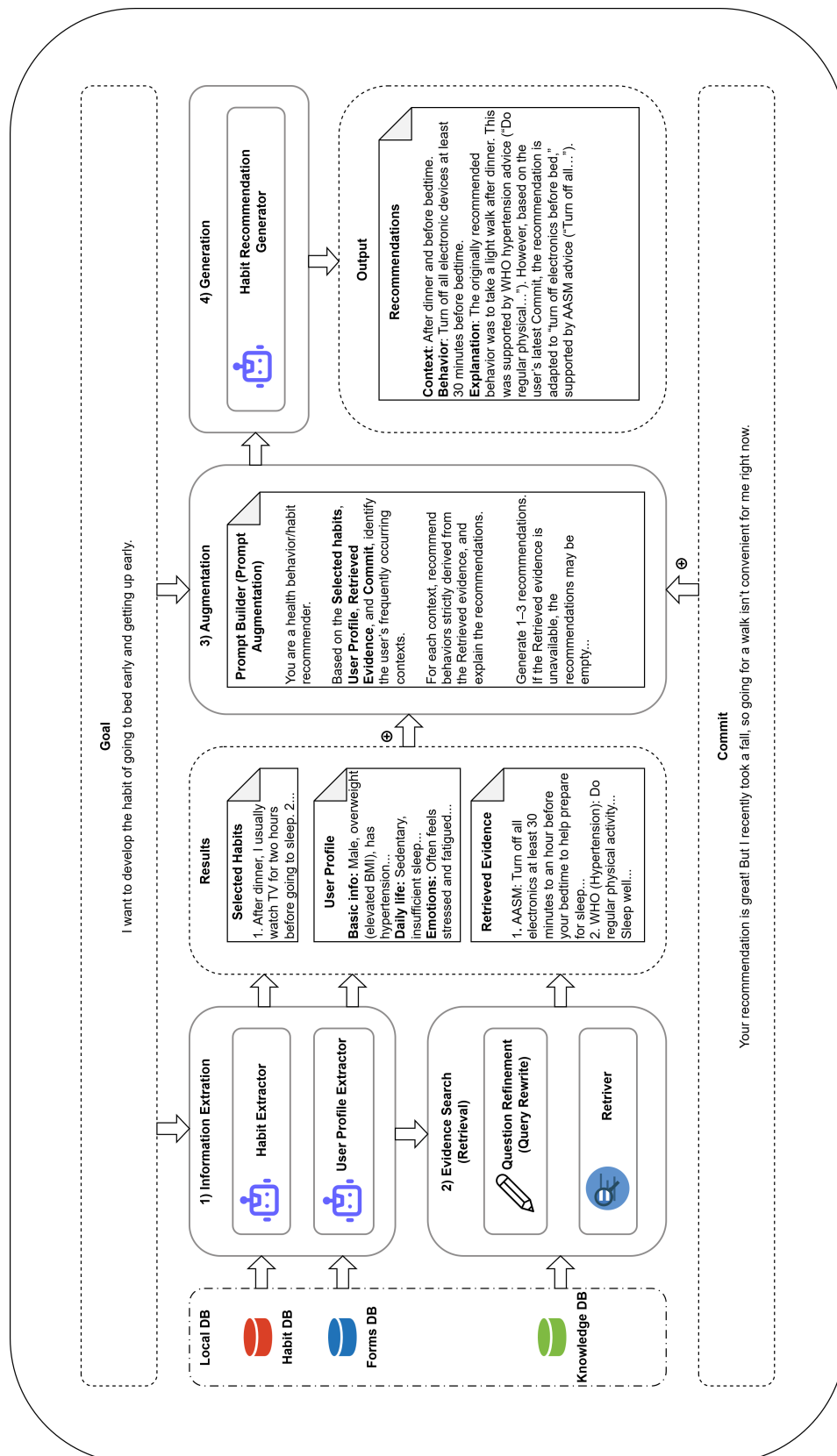


Figure 13: Conceptual Model of the Habit Recommender System

As shown in Figure 13, the Conceptual Model of the Habit Recommender System illustrates the overall process from user and knowledge inputs to habit recommendation generation. This process consists of four main stages, namely Information Extraction, Evidence Search (Retrieval), Augmentation, and Generation. First, the system extracts user-donated habit information and the user profile from the Local DB. Next, the system retrieves external evidence relevant to the current recommendation task from the Knowledge DB. Then, the Selected Habits, User Profile, Retrieved Evidence, the user's current Goal, and the most recent Commit are jointly incorporated into the Augmentation stage to construct the input for recommendation generation. Finally, based on this combined information, the system generates concrete habit recommendations together with their corresponding explanations. In addition, users can provide feedback on the recommendation results in the Commit stage, thereby forming a continuously iterative feedback loop.

4.3.4 Modular Structure of the Prototype

A module refers to a relatively independent unit within the prototype system that is organized around a specific functional task and design decision. Each module fulfills a distinct system role, has relatively clear input and output boundaries, and, through its coordination with other modules, jointly supports the realization of the overall workflow (Parnas, 1972; Bass et al., 2022; IEEE, 2009). In the present study, modules are not only structural units of the prototype system, but also constitute the implementation-level concretization of the DDs. The following sections are organized around the three core workflows and explain the functional role of each module, its input–output relationships, and its correspondence to the relevant DDs.

Module Name	Corresponding DDs
M1.1 Habit Classification	DD2–3, DD5, DD24
M1.2 Context Extraction	DD2–5, DD24
M1.3 Ontology Mapping	DD4, DD17–18, DD21
M1: Habit Donation	DD1–5, DD14, DD17, DD21
M2: Forms Management	DD6–7, DD10, DD14, DD21
M3.1 Habit Extractor	DD12, DD14, DD19, DD24
M3.2 User Profile Extractor	DD8–9, DD14, DD24
M3.3 Retrieval	DD13, DD16, DD18, DD21–22
M3.4 Habit Recommendation Generator	DD16, DD19, DD23
M3.5 Feedback Collector	DD11
M3: Habit Recommendation	DD8–9, DD11–14, DD15–16, DD19–24

Table 5: Mapping between system modules and corresponding DDs

Table 5 provides an overview of how the system modules implement the corresponding DDs.

4.3.4.1 Workflow 1

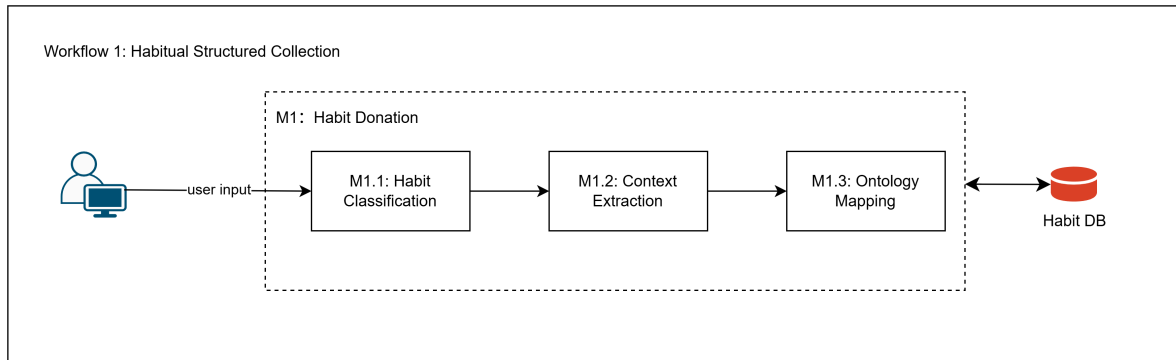


Figure 14: Workflow 1: Habitual Structured Collection

The role of the M1.1 Habit Classification module is to determine whether the user input describes a habit. Therefore, its input consists of the user-provided sentence, the language used for the habit, and a UUID, which is generated differently for each invocation. Its output includes all input information, the judgment of whether the input is a habit, the confidence of this judgment as provided by the LLM, and metadata about the model used, such as the provider and model name. A caching mechanism is introduced in this module in order to reuse previously computed classification results for identical inputs under identical LLM-related parameter settings.

The role of M1.2 Context Extraction is to identify and extract the phrases corresponding to habit-related context labels from the habit sentence entered by the user, including TIME, PHYSICAL SETTING, PRIOR BEHAVIOR, OTHER PEOPLE, INTERNAL STATE, BEHAVIOR, and REASONING, thereby providing structured input for the subsequent M1.3 Ontology Mapping. The input of M1.2 Context Extraction consists of the habit sentence, the language, and the UUID. Its output includes the input information, the phrases extracted for each context label, the confidence provided by the LLM, and metadata about the model used, such as the provider and model name. A caching mechanism is introduced in this module in order to reuse previously extracted context results for identical inputs under identical LLM-related parameter settings.

M1.3 Ontology Mapping is responsible for mapping the contextual phrases extracted by the M1.2 Context Extraction module to the corresponding concepts in BCIO. The input to this module consists of the habit sentence, the language, the UUID, and the phrases extracted under each contextual label. In addition, the mapping process incorporates user-adjustable parameter

settings, such as the similarity threshold and the number of top candidate matches. The output includes the original input information, as well as the corresponding BCIO label, IRI, and similarity score for each phrase, together with the applied mapping parameters, such as the similarity threshold and the number of top candidate matches.

As shown in Figure 14, M1: Habit Donation is in fact an orchestration module that implements the entire Workflow 1. More specifically, it connects the three submodules of the Habit Donation module, namely M1.1 Habit Classification, M1.2 Context Extraction, and M1.3 Ontology Mapping, and stores the outputs of all modules in the Habit DB. The module takes the user-provided habit text and the language as input. When the input is not classified as a habit, the output mainly contains the habit classification result and its confidence score. When the input is classified as a habit, the output further includes the context extraction results, the BCIO mapping results, the corresponding mapping parameters, and LLM metadata. This module adopts a dual-storage strategy. On the one hand, the latest results of the three submodules are stored separately in the corresponding collections of the Habit DB, using the sentence content itself as a unique key, so that only the most recent processing result for the same sentence is retained. On the other hand, the complete result of each M1 invocation is written to a separate history collection, using the UUID as a unique key, so that all historical records are preserved.

4.3.4.2 Workflow 2

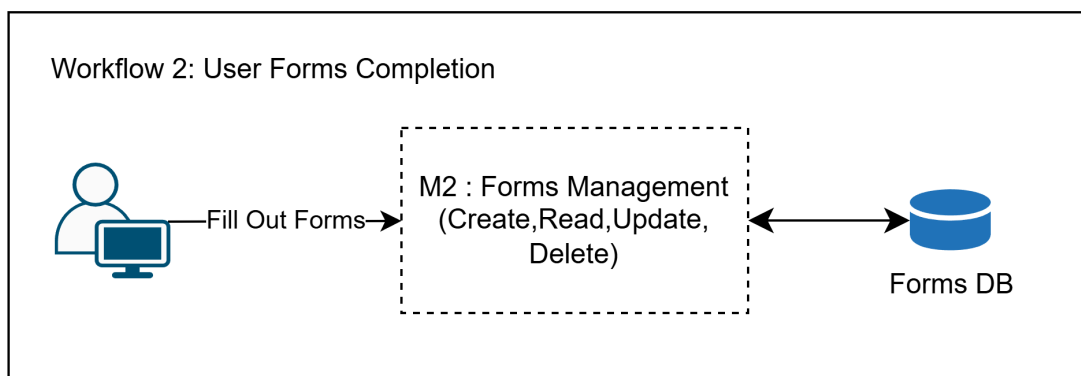


Figure 15: Workflow 2: User Forms Completion

As shown in Figure 15, M2: Forms Management is in fact also an orchestration module that implements the entire Workflow 2. More specifically, this module handles the process in which users fill in forms, supports the creation, reading, updating, and deletion of form data, and interacts with the Forms DB. In order to fully characterize the user, users are required to

complete three types of forms here: the Basis form (designed by the author to collect basic information and information related to NCDs), the Simple Lifestyle Indicator Questionnaire (SLIQ), which is used to collect information related to the user's daily lifestyle, such as diet, physical activity, smoking, alcohol consumption, and stress (Godwin et al., 2013), and the RAND 36-Item Health Survey (RAND-36), which is used to assess the user's health-related quality of life and overall health status (Hays et al., 1993). The information entered by the user in the three types of forms constitutes the input to this module. The latest version of each of the three forms is stored in the corresponding collection in the Forms DB and presented to the user for viewing and updating. At the same time, each modification of a form is written to a separate history collection in the Forms DB in order to preserve the complete historical records. The output of the module includes the form name, creation time, update time, and all information filled in by the user.

4.3.4.3 Workflow 3

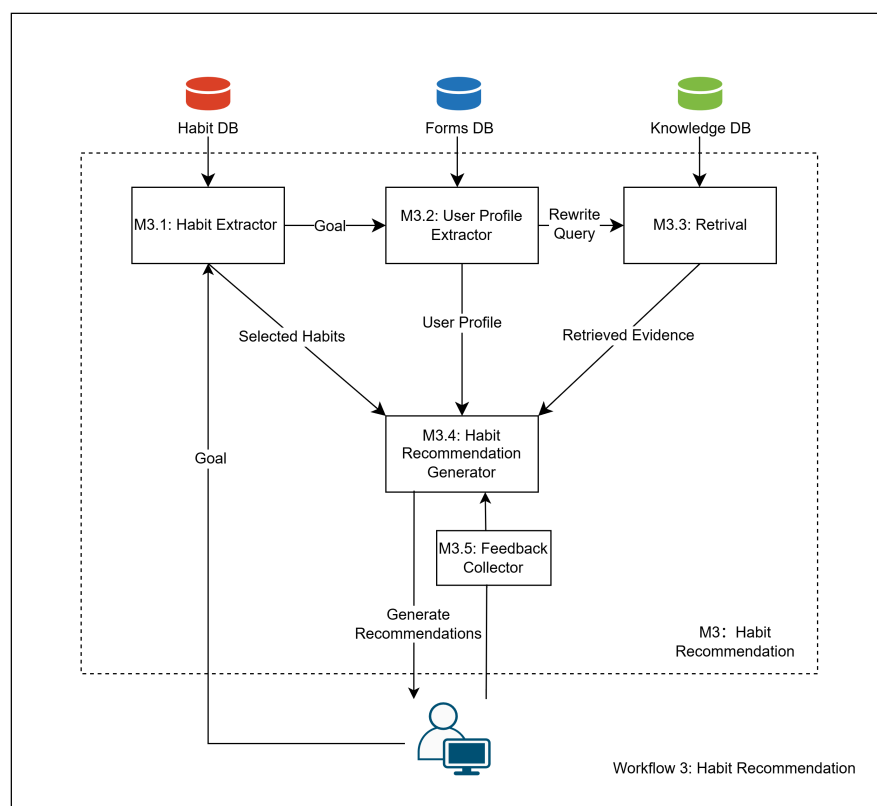


Figure 16: Workflow 3: Habit Recommendation

M3.1 Habit Extractor is responsible for identifying the habits in the Habit DB that are relevant to the user's current goal, thereby preventing too many irrelevant habits from being injected into the M3.4 Habit Recommendation Generator module. Accordingly, its input consists of the user-provided goal and the UUID generated to identify each invocation. Its output includes the original goal, the UUID, the selected habits (together with the importance score and reason provided by the LLM), and LLM metadata. The selected habits are compared with the habits stored in the database in order to prevent the LLM from hallucinating non-existent habits. A caching mechanism is introduced in this module to reuse habit selection results under the same goal, the same Habit DB, and the same parameter settings.

M3.2 User Profile Extractor is responsible for constructing a user profile relevant to the current recommendation task based on the user's goal and the form information stored in the Forms DB. Accordingly, its input consists of the goal passed from M3.1 and the UUID. Its output includes a detailed user profile, which is used both for presentation to the user and as an input to the M3.4 Habit Recommendation Generator module, as well as a simpler user profile for the subsequent retrieval process, namely the rewritten query. The output also includes LLM metadata. A caching mechanism is introduced in this module to reuse user profile generation results under the same goal, the same Forms DB, and the same parameter settings.

M3.3 Retrieval is responsible for retrieving evidence relevant to the current recommendation task from the Knowledge DB. Accordingly, its input consists of the rewritten query generated by M3.2 and the UUID. The documents in the Knowledge DB are first split into chunks and then stored in a vector database. The rewritten query is then compared with the stored vectors in terms of similarity in order to select the semantically most relevant retrieved evidence as the output. Each retrieved piece of evidence includes the original text, the page number, the document title, the document domain, and the similarity score. These evidence items are subsequently passed to the M3.4 Habit Recommendation Generator module. The output also includes retrieval-related parameters, such as the similarity threshold and the maximum number of retrieved evidence items to be returned. In addition, the M3.3 Retrieval module also handles updates to the Knowledge DB. That is, whenever the contents of the Knowledge DB change, the M3.3 Retrieval module needs to respond appropriately, for example by re-chunking the documents and updating the vector database.

M3.4 Habit Recommendation Generator is responsible for generating habit recommendations by integrating the selected habits, the extracted user profile, the retrieved evidence, and the collected feedback. Accordingly, its input mainly consists of the selected habits from M3.1, the user profile from M3.2, the retrieved evidence from M3.3, and the feedback information from M3.5. These pieces of information are combined into a prompt, on the basis of which the

recommendations are generated. Its output includes the habit recommendations provided to the user. Each recommendation includes the context in which the user can perform the recommended behavior, the specific behavior itself, and an explanation that helps the user understand why the LLM produced this recommendation. A caching mechanism is introduced in this module in order to reuse recommendation results for fully identical inputs under identical parameter settings.

M3.5 Feedback Collector is responsible for collecting user feedback on the generated recommendations. Accordingly, its input consists of the feedback provided by the user in response to the generated recommendations and the UUID. Its output is the input itself, and this information is passed to M3.4 in order to support subsequent recommendation generation.

As shown in Figure 16, M3: Habit Recommendation is in fact an orchestration module that implements the entire Workflow 3. More specifically, it connects the five submodules M3.1 Habit Extractor, M3.2 User Profile Extractor, M3.3 Retrieval, M3.4 Habit Recommendation Generator, and M3.5 Feedback Collector, while interacting with the Habit DB, the Forms DB, and the Knowledge DB. The workflow is triggered by the user-provided goal. Based on this goal, the system first selects relevant habits from the Habit DB, extracts a user profile from the Forms DB, retrieves supporting evidence from the Knowledge DB, and finally generates habit recommendations for the user. In addition, user feedback on the generated recommendations is collected and incorporated into the workflow, thereby forming an iterative recommendation loop.

4.3.5 Prototype Implementation

Based on the conceptual model shown in Figure 13, as well as the module diagrams of Workflow 1 (Figure 14), Workflow 2 (Figure 15), and Workflow 3 (Figure 16), a prototype of the Habit Recommender System was implemented as the design artifact. The detailed implementation code is provided in the GitHub repository (Hua, 2026).

The Habit Recommender System was designed as a three-layer architecture. The bottom layer is the Backend API Layer, which provides the API endpoints for the submodules shown in the module diagrams. The middle layer is the Backend Orchestration Layer, which invokes the APIs of the Backend API Layer to implement the higher-level modules, such as M1, M2, and M3, while also exposing selected API endpoints for direct interaction with the frontend. Table 6 shows the mapping between the system modules and the API endpoints.

Module Name	API Endpoint	Layer	
M1.1 Habit Classification	POST/classify_habit	Backend API Layer	
M1.2 Context Extraction	POST/classify_context	Backend API Layer	
M1.3 Ontology Mapping	POST/map	Backend API Layer	
M1: Habit Donation	POST/ingest	Backend	Orchestration Layer
M2: Forms Management	PUT/profile/{profile_uuid}/latest and GET/profile/{profile_uuid}/latest	Backend	Orchestration Layer
M3.1 Habit Extractor	POST/habit_db/select	Backend API Layer	
M3.2 User Profile Extractor	POST/profiles/build	Backend API Layer	
M3.3 Retrieval	POST/kb/query	Backend API Layer	
M3.4 Habit Recommendation Generator	POST/recommend	Backend API Layer	
M3.5 Feedback Collector	POST/recommend/comment	Backend	Orchestration Layer
M3: Habit Recommendation	POST/recommend	Backend	Orchestration Layer

Table 6: Mapping between system modules, API endpoints, and architectural layers

The Frontend Layer implements a DD-based user interface by directly calling the APIs provided by the HHH backend layer. It consists of five functional pages: Donate Habit, Manage Habits, Health Profile, Get Recommendation, and Manage Recommendations. As shown in Figures 17, Donate Habit and Manage Habits implement Workflow 1. As shown in Figure 18, Health Profile implements Workflow 2. As shown in Figures 19, Get Recommendation and Manage Recommendations implement Workflow 3.

(a) Donate Habit

(b) Manage Habits

Figure 17: User interfaces supporting Workflow 1, including habit donation and habit management

Habit Recommender System

Donate Habit Manage Habits **Health Profile** Get Recommendation Manage Recommendations

Health Profile Basic SLIQ RAND-36

Please honestly complete the three forms **Basic** / **SLIQ** / **RAND-36**. You can update your answers at any time. The forms will be stored in our local database and used later to generate recommendations.

Basic Health Profile

The Basic Health Profile is an author-developed questionnaire created for this prototype to collect additional self-reported information and enrich the user profile.

Age *

25

Sex / Gender *

Male

Height (cm) *

172

Weight (kg) *

76

Physical limitations *

(a) Health Profile

Figure 18: User interface supporting Workflow 2 for health profile management

Habit Recommender System

Donate Habit Manage Habits **Health Profile** **Get Recommendation** Manage Recommendations

Get recommendations

Enter a specific behavior you want to improve, or a broader goal. Based on your existing form data, daily habits, and personal knowledge base, the system will generate actionable habit suggestions tailored to your everyday context.

Goal:

Enter your goal first – we'll generate personalized recommendations for you.

For example:
I want to improve my sleep quality. I often scroll on my phone in bed, fall asleep slowly, and struggle to get up in the morning. I'd like to build a better bedtime routine.

Generate recommendations

Go to management

Habit Recommender System

Donate Habit Manage Habits **Health Profile** **Get Recommendation** **Manage Recommendations**

Recommendation History

You can review an overview of your personalized recommendations and all the information used to generate them on this page.

Refresh

Total: 14 Items per page: 20 Prev Page 1 / 1 Next

goal I want to sit less and be more active every day. Expand Labels

goal 我想长高 Expand Labels

goal My goal is to build a healthier bedtime routine. Expand Labels

goal 我想变瘦 Expand Labels

goal 我想变健康 Expand Labels

goal I want to lower my blood pressure by improving my diet and daily routine. Expand Labels

goal 我想改善饮食 Expand Labels

goal 我想早睡 Expand Labels

goal 我想更自律 Expand Labels

(a) Get Recommendation

(b) Manage Recommendations

Figure 19: User interfaces supporting Workflow 3, including recommendation generation and recommendation management

The main programming language of the entire project is Python. The frontend technology

stack uses Vue 3 ([Vue.js, 2026](#)), while the backend technology stack uses FastAPI ([FastAPI Team, 2025](#)). FastAPI was chosen because it relies on two core libraries, Starlette and Pydantic. Starlette supports asynchronous processing, while Pydantic supports data validation. The former can support possible large-scale concurrent user requests in the future, whereas the latter can validate and transform the outputs extracted by LLMs. Compared with Django, FastAPI is more lightweight.

For the operational database, MongoDB was chosen instead of a traditional relational database. The main reason is that MongoDB does not enforce fixed columns, which makes it convenient to extend the schema when new context information needs to be added. In addition, the data is naturally JSON-like, which makes reading and writing simple. Three independent databases were defined according to the different functions of the three workflows. HabitDB serves Workflow 1 and is used to store the latest output data of M1.1, M1.2, and M1.3, with the habit sentence itself as the primary key. At the same time, there is also a history collection that stores the output of each call to M1, with UUID as the primary key of the history collection. UserProfiles serves Workflow 2. Three collections are used to store the latest data of the user's three forms, with the form name as the primary key. There is also a history collection that stores all information from each update of the three forms, with UUID as the primary key of the history collection. RecommendationsDB serves Workflow 3. Five collections are used to store the latest data of modules M3.1 to M3.5, with the user's goal as the primary key. There is also a history collection that stores the output of each call to M3, with UUID as the primary key of the history collection. In this way, for each module and the API behind it, there are collections that store both the latest information and the historical information.

For the embedding model, M3-Embedding (BGE-M3) was used both in ontology mapping and in the subsequent knowledge base retrieval. The reason is that it supports dense, sparse, and multi-vector retrieval within a unified model and covers multilingual as well as long-document input, which is in line with DD2 ([Chen et al., 2025](#)). The difference is that, in ontology mapping, the labels and definitions of the ontology classes in the OWL file were used to construct dense and sparse vectors, and the similarity was obtained through linear weighted fusion. In this way, dense representations can be used to capture semantically similar relations, while sparse representations can preserve keyword matching signals, thereby improving the robustness and accuracy of the mapping results ([Chen et al., 2025](#); [Wang et al., 2021b](#)). In knowledge base retrieval, by contrast, dense vectors, that is, what are commonly referred to as vectors in everyday use, were used to calculate similarity.

For the vector database, Milvus was chosen because, as a purpose-built database for vector data management, it can efficiently support large-scale vector storage and similarity retrieval

(Wang et al., 2021a).

For the construction of the local knowledge base, the local knowledge base is located in the `kb` folder. The contents of this folder can be modified by the user, but the predefined structure must be followed. At present, the local knowledge base only supports PDF files, and the file names are preferably written with underscores instead of spaces. The names of the subfolders under `kb` represent domain names, so that users can classify documents by domain. The `_meta` folder under `kb` contains the relevant information for each PDF and its corresponding chunks. The user can update the local knowledge base at any time. Whenever the M3.3 Retrieval module is invoked, it first checks whether the local knowledge base has been updated and then adapts accordingly, for example by re-chunking the documents and updating the vector database. The structure of the `kb` folder is shown in Listing 1.

Listing 1: Example directory structure of the local knowledge base

```
kb/
|-- _meta/
|   |-- _kb_state.json
|   |-- <doc_id_1>.json
|   |-- <doc_id_2>.json
|   |-- <doc_id_3>.json
|   |-- <doc_id_4>.json
|   |-- <doc_id_5>.json
|-- hypertension/
|   |-- 2023_ESH_Guidelines_for_the_Management_of_Arterial_Hypertension.pdf
|-- metabolic/
|   |-- Type_2_Diabetes_Prevention_in_People_at_High_Risk.pdf
|-- musculoskeletal/
|   |-- Management_of_Osteoarthritis_of_the_Knee_Non_Arthroplasty_Evidence_Based_Clinical_Practice_Guideline.pdf
|-- sleep/
|   |-- Improvements_in_Sleep_Quality_and_Fatigue_Are_Associated_with_Improvements_in_Functional_Recovery_Following_Hospitalization_in_Older_Adults.pdf
|   |-- The_Effect_of_Physical_Activity_on_Sleep_Quality_and_Sleep_Disorder_A_Systematic_Review.pdf
```

For LLM calls, this thesis used OpenAI API keys and ScaDS.AI API keys. In the prompting stage, few-shot prompting was used only in `POST/classify_habit`, while one-shot

prompting was used only in `POST/classify_context`. One-shot means that, in addition to the actual input, one example is provided to the model to show how the task should be completed. Few-shot, by contrast, provides multiple examples (Schulhoff et al., 2025). The reason is that, in the course paper, it was found that most LLMs performed better on the few-shot habit classification task and the one-shot context extraction task. Apart from this, in other LLM calls, the prompt was mainly structured by clearly specifying the goal of the LLM, the task, the exact input format and output format, as well as relevant notes and constraints. This kind of structured prompt design is also consistent with recommendations in the existing prompt engineering literature (Hewing and Leinhos, 2024; Mao et al., 2025; White et al., 2023).

The prompt design of the M3.4 Habit Recommendation Generator adopts a constrained fallback strategy. Specifically, when the RAG results do not provide directly executable behavioral recommendations but contain clear, relevant, and normatively supported evidence or behavioral directions, the system is allowed to cautiously instantiate this direction into a reasonable behavioral recommendation and explain the inferential basis in the explanation. Conversely, if the RAG results contain neither a concrete executable behavior nor relevant evidence sufficient to support such an inference, the module returns an empty recommendation list (Rajpurkar et al., 2018). This design aims to balance recommendation usefulness and output accuracy by avoiding unsupported recommendations when evidence is insufficient, while also preventing the system from abandoning a recommendation too early when a reasonable evidence-based direction is available.

For caching, Redis was adopted mainly because, as an in-memory data store, it supports fast reading and writing of cached results, which helps reduce the response latency and operational costs caused by repeated calls to LLMs (Regmi and Pun, 2024). Redis was used in the M1.1 Habit Classification, M1.2 Context Extraction, M3.1 Habit Extractor, M3.2 User Profile Extractor, and M3.4 Habit Recommendation Generator modules. To simplify the system implementation, the TTL of the cache was uniformly set to 600 seconds for all these modules.

For deployment, the Backend API Layer, the Backend Orchestration Layer, and the Frontend Layer were all dockerized, thereby ensuring that the system components can be deployed and run in a relatively independent and consistent manner.

4.4 Demonstration

Peffer et al. (2007) propose that, in the demonstration stage, it is necessary to show how the artifact can be used to solve one or more problem instances. A persona is a prototypical and concretized representation of a certain type of target user. It is usually given a name and a face and is described around its needs, goals, and tasks. Personas were originally defined to enable design

teams to discuss a concrete user image throughout the design process ([Blomquist and Arvola, 2002](#)). [Pruitt and Grudin \(2003\)](#) further extend the role of personas as an information carrier and communication medium throughout the product development process, so that personas not only support design discussions, but are also used in development and testing activities such as demos, bug bashes, and QA testing. Therefore, personas can concretize otherwise abstract target users and problems, and can thus be used in the demonstration stage of DSR.

Regarding how personas can be constructed, existing literature suggests that there is not one universal method for constructing personas; instead, different methods organize personas according to project goals and domain knowledge ([Jansen et al., 2017](#)). Traditional persona creation methods include qualitative persona creation, which is usually based on user interviews, focus groups, ethnographic observation, and open or axial coding. Based on real user data, typical user representations are formed through manual interpretation and induction. Another approach is quantitative persona creation, which is based on online data and user behavior logs and applies methods such as clustering or regression. In addition, mixed-methods persona creation combines both approaches ([Qatar Computing Research Institute et al., 2022](#)). However, considering the required effort, project scope, and feasibility, literature-informed persona construction was adopted. Previous studies have indicated that, in situations where direct access to the target group is difficult or sufficient empirical participation cannot be conducted, personas can still be constructed based on relevant literature ([Tariq et al., 2012](#); [Veldsman and Coetzee, 2022](#)).

First, it must be clarified that the personas were created for the demonstration and evaluation stages. The purpose of creating the personas is to examine whether all or part of the meta-requirements are satisfied ([Peffers et al., 2007](#); [Venable et al., 2016](#)). An overview of the three personas and how information was mapped from the meta-requirements is provided in Figure 20. As can be seen, Personas 1, 2, and 3 all cover all meta-requirements, but each persona has a different emphasis.

Second, the creation of the persona components and the specific persona content was based on existing literature, such as literature related to NCDs and studies that used similar persona components in the health domain ([Tariq et al., 2012](#); [Veldsman and Coetzee, 2022](#)).

In the creation of the persona components, no single template was directly adopted. Instead, the components were adjusted purposefully based on existing persona components and the intended evaluation purpose. Each persona includes (1) a basic profile, drawing on personal background factors and covering basic user information, such as age, gender, and certain behavioural habits ([Schäfer et al., 2019](#)); (2) the health situation, which may include aspects of physical, mental, and cognitive health ([Guan et al., 2021](#)); (3) goals, motivations, and barriers, drawing on persona research that highlights goals, tasks, and motivations, as well as on recent

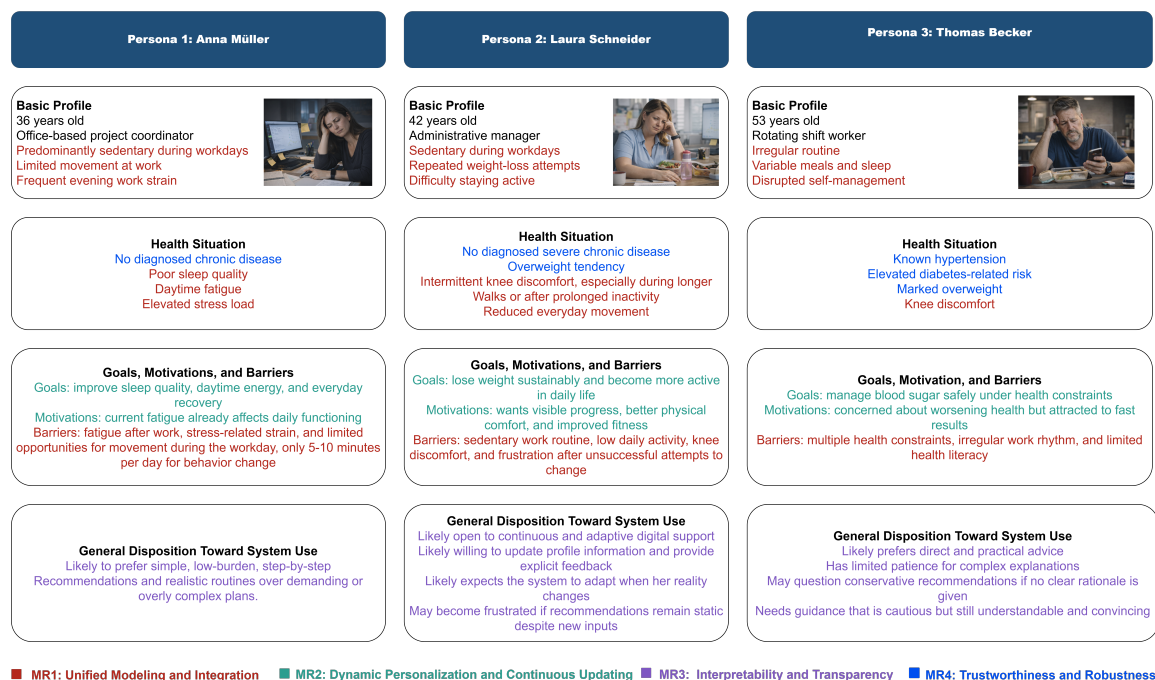


Figure 20: Three evaluation-oriented personas and their color-coded mapping to the meta-requirements

behaviour-change-oriented persona work that explicitly incorporates barriers (McMullin et al., 2021; -ul-Ain et al., 2024); and (4) a general disposition towards system use, reflecting users' thoughts, technology use, and technological skills (Bartels et al., 2023).

The creation of the specific persona content was mainly based on existing literature, while also considering the purpose of the evaluation. On this basis, three personas were further summarized and derived. For example, sedentary behaviour and physical inactivity are relevant modifiable behavioural risk factors in the prevention of NCDs (Budreviciute et al., 2020). In addition, device-measured evidence shows that office workers are among the most sedentary occupational groups and accumulate substantial sitting time during work (Prince et al., 2019a). Therefore, Anna Müller was designed as an office-based adult with a predominantly sedentary daily work routine. The specific derivation process is documented in the `PersonasCreation` worksheet of the `Personas.xlsx` file in the GitHub repository (Hua, 2026). A concrete overview of the three personas and the literature used for their construction is shown in Table 7.

Table 7: Overview of the Three Personas and the Literature Used for Their Construction

Personas	Basic Profile (Schäfer et al., 2019)	Health Situation (Guan et al., 2021)	Goals, Motivations, and Barriers (McMullin et al., 2021; ul-Ain et al., 2024)	General Disposition Toward System Use (Bartels et al., 2023)
Anna Müller	36 years old; office-based project coordinator; predominantly sedentary during workdays; limited movement at work; frequent evening work strain (Prince et al., 2019a).	No diagnosed chronic disease; poor sleep quality; daytime fatigue; elevated stress load (Mao et al., 2023; Prince et al., 2019a).	Goals: improve sleep quality, daytime energy, and everyday recovery. Motivations: current fatigue already affects daily functioning. Barriers: fatigue after work, stress-related strain, and limited opportunities for movement during the workday (Mao et al., 2023; Prince et al., 2019a); only 5–10 minutes per day for behavior change.	Likely to prefer simple, low-burden, step-by-step recommendations and realistic routines over demanding or overly complex plans (Mao et al., 2023; Prince et al., 2019a).
Laura Schneider	42 years old; administrative manager; sedentary during workdays; repeated weight-loss attempts; difficulty staying active (Prince et al., 2019b; Zheng and Chen, 2015).	No diagnosed severe chronic disease; overweight tendency; intermittent knee discomfort, especially during longer walks or after prolonged inactivity; reduced everyday movement (Prince et al., 2019b; Zheng and Chen, 2015).	Goal: lose weight sustainably and become more active in daily life. Motivations: wants visible progress, better physical comfort, and improved fitness. Barriers: sedentary work routine, low daily activity, knee discomfort, and frustration after unsuccessful attempts to change (Prince et al., 2019b; Zheng and Chen, 2015).	Likely open to continuous and adaptive digital support; likely willing to update profile information and provide explicit feedback; likely expects the system to adapt when her reality changes; may become frustrated if recommendations remain static despite new inputs (Prince et al., 2019b; Zheng and Chen, 2015).
Thomas Becker	53 years old; rotating shift worker; irregular routine; variable meals and sleep; disrupted self-management (Boini et al., 2022).	Known hypertension; elevated diabetes-related risk; marked overweight; knee discomfort (Boini et al., 2022; Zheng and Chen, 2015).	Goal: manage blood sugar safely under health constraints. Motivations: concerned about worsening health but attracted to fast results. Barriers: multiple health constraints, irregular work rhythm, and limited health literacy (Boini et al., 2022; World Health Organization, 2025a).	Likely prefers direct and practical advice; has limited patience for complex explanations; may question conservative recommendations if no clear rationale is given; needs guidance that is cautious but still understandable and convincing (World Health Organization, 2025a; Boini et al., 2022).

Anna Müller represents a persona design oriented toward early prevention. She mainly focuses on office-based sedentary behaviour, work stress, reduced sleep quality, and daytime fatigue, and is used to examine whether the system can provide feasible recommendations for users who are still in a modifiable stage ([Prince et al., 2019a](#); [Mao et al., 2023](#)).

Laura Schneider represents a persona design oriented toward continuous behavioural adjustment and dynamic adaptation. She differs from Anna Müller in that she has already been facing long-term difficulties with weight management, repeated weight-loss attempts, and knee discomfort. She is used to examine whether the system can provide sufficiently effective recommendations that support long-term behaviour change ([Prince et al., 2019b](#); [Zheng and Chen, 2015](#)).

Thomas Becker represents a persona design oriented toward complex constraints. Due to his shift work and limited health literacy, he is mainly used to examine whether the system can provide reliable recommendations that meet user needs in a high-risk context ([Boini et al., 2022](#); [Zheng and Chen, 2015](#); [World Health Organization, 2025a](#)).

To avoid repetition, the persona Anna Müller is used as the example in the demonstration stage. According to the evaluation episodes defined in the following evaluation section, Anna Müller first enters three habit sentences, completes the form data and modifies it twice, and then enters her goal: “I want to improve my sleep quality.” The concrete input examples are documented in the `EvaluationDesign` worksheet of the `Evaluation.xlsx` file in the GitHub repository ([Hua, 2026](#)).

goal

I want to improve my sleep quality.

Collapse Labels

habit recommendations

provider: scads

model: openai/gpt-oss-120b

Collapse Labels

The recommendations were generated successfully. However, if the results are empty or do not meet your expectations, please consider whether you entered the correct goal, whether all forms have been completed correctly, whether the documents stored in the local knowledge base can support your goal, and whether you have provided a sufficient number of high-quality habits. Higher quality inputs usually lead to better recommendation results.

in the evening after work, when you feel work-related strain

Collapse Labels

behavior:

Spend 5-10 minutes doing gentle stretching or deep-breathing (a light mind-body routine) to lower stress before bedtime.

context

in the evening after work, when you feel work-related strain

behavior

Spend 5-10 minutes doing gentle stretching or deep-breathing (a light mind-body routine) to lower stress before bedtime.

explanation

- **Context choice**: Your selected habit notes "I always experience work strain in the evening," indicating that evening is a recurring stressful moment you already recognize.
- **Behavior source**: The RAG material states that "physical activity reduces stress, which is a typical impediment to falling and staying asleep" and that "light exercises of 10 minutes per day improve sleep quality." It also highlights that "mind-body physical exercises like yoga improve sleep outcomes." Combining these points, a brief low-intensity stretch/breathing routine is a concrete, evidence-based action.
- **Fit to your situation**: The routine lasts only 5-10 minutes, matching your daily time budget, and is low-burden, aligning with your preference for simple, realistic changes.
- **Difference from current habits**: You currently have limited movement at work and sit most of the day; this evening movement introduces new activity rather than repeating existing sedentary behavior.
- **How it helps your goal**: Reducing evening stress through gentle movement lowers pre-sleep arousal, shortens sleep latency, and promotes relaxation, thereby improving overall sleep quality.

RAG retrieval results

provider: scads

model: openai/gpt-oss-120b

top_n: 35

threshold: 0.65

Expand Labels

selected habits

provider: scads

model: openai/gpt-oss-120b

Expand Labels

bilded profiles

provider: scads

model: openai/gpt-oss-120b

Expand Labels

feedback

Expand Labels

Figure 21: Demonstration example of the recommendation result for Persona 1

The specific parameters and recommendation results can be seen in Figure 21. The context was inferred from the donated habit sentence "I always experience work strain in the evening", and the recommendation closely follows the evidence retrieved, namely that "physical activity reduces stress, which is a typical impediment to falling and staying asleep" and that "light exercises of 10 minutes per day improve sleep quality." More specifically, the recommendation is also a reasonable inference based on the literature. For example, although the original text only states that mind-body physical exercises such as yoga can improve sleep, the recommended behaviour, namely gentle stretching or deep breathing, can be regarded as a concrete implementation of

mind-body physical exercises. In addition, the duration of 5–10 minutes was determined by jointly considering the literature support and the user profile. The recommendation also matches the user’s goal: “I want to improve my sleep quality.”

4.5 Evaluation

To ensure methodological rigor, the evaluation was structured according to the FEDS framework (Peffer et al., 2007; Venable et al., 2016). The FEDS evaluation design process consists of four steps: (1) explicate the goals of the evaluation, (2) choose the evaluation strategy or strategies, (3) determine the properties to evaluate, and (4) design the individual evaluation episodes. As shown in Figure 22, the results of each step are presented. Subsequently, the results of the preliminary evaluation of the Habit Recommender System are presented in three concept matrices, namely Table 8, Table 9, and Table 10.

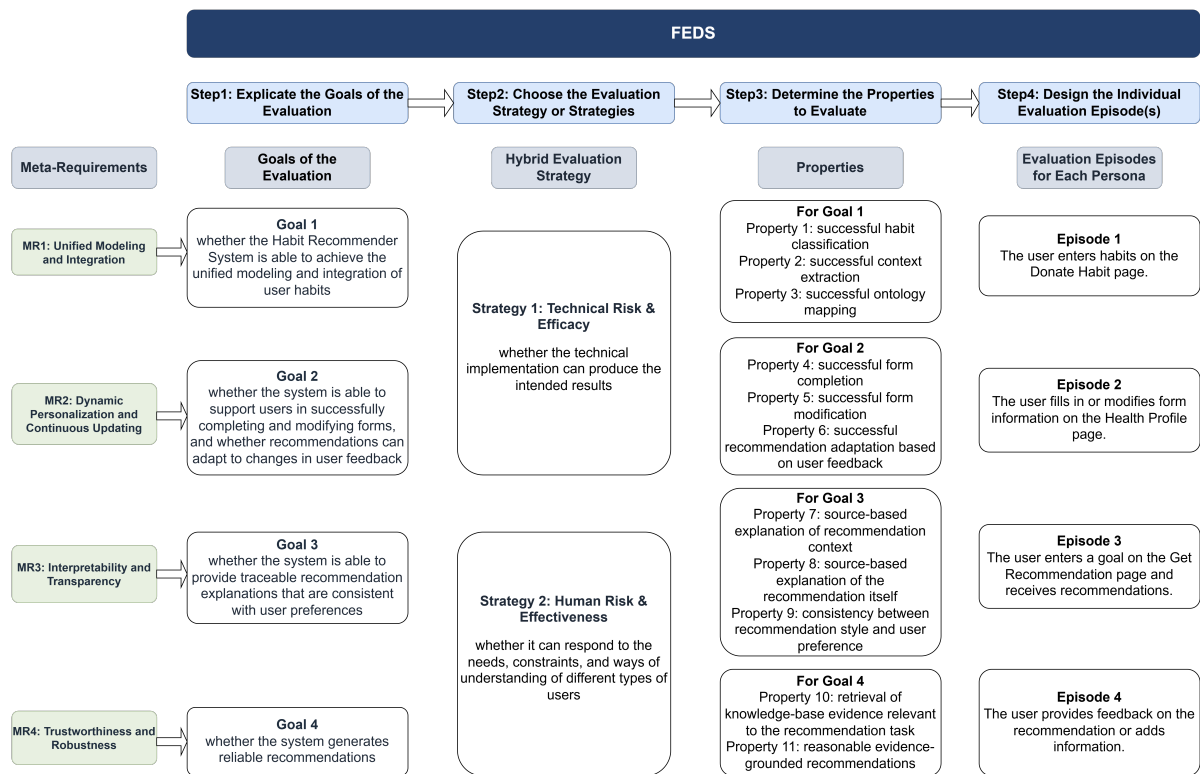


Figure 22: Adapted from the FEDS evaluation design process (Venable et al., 2016) and applied to the preliminary evaluation of the Habit Recommender System

Peffer et al. (2007) define the evaluation stage as the observation and measurement of the extent to which an artifact supports problem solving, and point out that this stage requires comparing the previously defined solution objectives with the actual results observed from

the artifact. In the present research, the solution objectives are expressed in the form of meta-requirements. Accordingly, the evaluation goals are derived first from the meta-requirements defined in the preceding sections.

Venable et al. (2016) argue that resource constraints should be taken into account when designing an evaluation. Therefore, given the limitations of time, resources, and the current stage of prototype development, the evaluation focuses on those aspects of the meta-requirements that can be verified, rather than attempting a complete, long-term, and comprehensive validation of all objectives under real deployment conditions. More specifically, the evaluation is structured around four meta-requirements: first, Goal 1 whether the Habit Recommender System is able to achieve the unified modeling and integration of user habits; second, Goal 2 whether the system is able to support users in successfully completing and modifying forms, and whether recommendations can adapt to changes in user feedback; third, Goal 3 whether the system is able to provide traceable recommendation explanations that are consistent with user preferences; and fourth, Goal 4 whether the system generates reliable recommendations.

Venable et al. (2016) define an evaluation strategy as a planned trajectory of evaluation that is suitable for the specific context of a DSR project. In this thesis, Strategy 1 Technical Risk & Efficacy refers to testing the system under relatively controlled conditions in order to examine whether it functions as intended by design. Strategy 2 Human Risk & Effectiveness, in contrast, concerns whether the system remains useful and acceptable when applied in real or near-real usage contexts involving actual users and practical problems.

Based on the goals outlined above, a hybrid evaluation strategy was adopted. On the one hand, for core modules such as habit classification, context extraction, ontology mapping, retrieval, and recommendation generation, the main focus of the evaluation is to verify whether the technical implementation can produce the intended results. In this respect, the evaluation is closer to the Technical Risk & Efficacy orientation in FEDS. On the other hand, since the artifact is ultimately intended for health recommendation scenarios involving real users, its effectiveness also depends on whether it can respond to the needs, constraints, and ways of understanding of different types of users. Therefore, Human Risk & Effectiveness is also incorporated. In practical terms, this hybrid evaluation strategy was operationalized through persona-based evaluation episodes.

According to Venable et al. (2016), the core of this step is to answer the question of “what to evaluate,” that is, to determine which characteristics, objectives, and requirements of the artifact should be selected as concrete evaluation objects by considering the purpose of the artifact itself, the context in which it is situated, as well as the evaluation goals and evaluation strategy that have already been clarified in the previous step. More specifically, with regard to Goal 1, the evaluation properties include Property 1 whether habit classification can be

performed successfully, Property 2 whether context extraction can be performed successfully, and Property 3 whether ontology mapping can be performed successfully. With regard to Goal 2, the evaluation properties include Property 4 whether form completion can be carried out successfully, Property 5 whether the modification process can be carried out successfully, and Property 6 whether recommendations can be adapted successfully in response to user feedback. With regard to Goal 3, the evaluation properties include Property 7, whether the context of the recommendation is accompanied by source-based explanations, Property 8 whether the recommendation itself is accompanied by source-based explanations, and Property 9 whether the recommendation style is consistent with the preferences provided by the user. With regard to Goal 4, the main focus at the current stage is to evaluate Property 10 whether the system can retrieve knowledge-base evidence relevant to the recommendation task and, on this basis, Property 11 whether it can generate reasonable evidence-grounded recommendations.

According to [Venable et al. \(2016\)](#), evaluation episodes refer to specific evaluation activities conducted within the overall evaluation strategy, each centered on a particular evaluation object and carried out by using a particular method. Early evaluation can be conducted with the help of persona descriptions, use-cases, concept models, and similar forms ([Kopenhagen et al., 2012](#)). Personas have also increasingly been used in prototype design and in the continuous iterative evaluation of recommender systems. Personas can not only support the examination of prototypes in relation to user contexts, but can also serve as a vehicle for qualitative evaluation in order to compare the suitability and effectiveness of system designs across different user scenarios ([Misztal-Radecka and Indurkha, 2020](#); [Dotan et al., 2009](#); [Thaker et al., 2024](#)).

Therefore, in the present research, personas are not used as substitutes for real users, but rather as contextualized vehicles for evaluation, through which abstract design goals can be translated into executable and comparable evaluation episodes. Anna Müller, Laura Schneider, and Thomas Becker represent three different types of scenarios, namely early preventive support, dynamic behavior adjustment, and high-risk management under complex constraints. For each persona, corresponding goals, habit inputs, form information, health constraints, and a shared knowledge base background were designed on the basis of the persona's existing attributes, and the system's outputs in classification, extraction, mapping, retrieval, recommendation, and explanation were observed in that scenario. In addition to the initial scenario, follow-up scenarios after user feedback were also included in order to examine whether the system is able to adapt on the basis of new inputs. In practical terms, the evaluation episodes in the present research can be divided into the following four types:

- Episode 1: the user enters habits on the Donate Habit page, through which the system's processing results for habit classification, context extraction, and ontology mapping are

evaluated.

- Episode 2: the user fills in or modifies form information on the Health Profile page, through which the system's support for user information input and updating is evaluated.
- Episode 3: the user enters a goal on the Get Recommendation page and receives a recommendation, through which the system's retrieval, recommendation, and explanation outputs under a given persona scenario are evaluated.
- Episode 4: the user provides feedback on the recommendation or supplies additional information, through which it is evaluated whether the system is able to update the recommendation effectively on the basis of new inputs.

The evaluation episodes were created based on the characteristics of the existing personas, as well as the previously derived evaluation properties and evaluation goals. The detailed derivation process is documented in the `EvaluationDesign` worksheet of the `Evaluation.xlsx` file in the GitHub repository ([Hua, 2026](#)).

To facilitate a unified evaluation, the knowledge base was designed as a shared database for the three personas. The selection of literature was based primarily on the personas' health situations, goals, and barriers. For Anna Müller, given her poor sleep quality, daytime fatigue, and elevated stress load, as well as her goals of improving sleep quality, increasing daytime energy, and promoting everyday recovery, literature related to sleep quality, fatigue recovery, and the relationship between physical activity and sleep was included ([Arentson-Lantz et al., 2022](#); [Alnawwar et al., 2023](#)). For Laura Schneider and Thomas Becker, given that both have weight-management problems as well as knee discomfort, a guideline on type 2 diabetes prevention in people at high risk and lifestyle intervention, as well as a guideline on the management of knee osteoarthritis, were included ([National Institute for Health and Care Excellence, 2017](#); [American Academy of Orthopaedic Surgeons, 2021](#)). Considering that Thomas Becker also has known hypertension and that his goal emphasizes managing blood sugar safely under health constraints, a hypertension management guideline was further included ([Mancia et al., 2023](#)).

Subsequently, the preliminary system evaluation was conducted according to the designed episodes. The specific evaluation results and the evaluation analysis are documented in the `EvaluationAllResults` worksheet of the `Evaluation.xlsx` file in the GitHub repository ([Hua, 2026](#)). The results were further summarized in the concept matrices shown in Table 8, Table 9, and Table 10.

Table 8: Concept Matrix Mapping Evaluation Goals to Episodes for Anna Müller

		Episode 1			Episode 2			Episode 3			Episode 4		
		Habit 1	Habit 2	Habit 3	Form Completion	Modification 1	Modification 2	Goal 1	Goal 2	Goal 3	Feed-back 1	Feed-back 2	Feed-back 3
Goal 1	Property 1	X	X	X									
	Property 2	X	X	X									
	Property 3	X	X	X									
Goal 2	Property 4				X								
	Property 5					X	X						
	Property 6										X	X	X
Goal 3	Property 7							X	X	X			
	Property 8							X	X	X			
	Property 9												
Goal 4	Property 10							X	X	X			
	Property 11							X		X			

Table 9: Concept Matrix Mapping Evaluation Goals to Episodes for Laura Schneider

		Episode 1			Episode 2			Episode 3			Episode 4		
		Habit 1	Habit 2	Habit 3	Form Completion	Modification 1	Modification 2	Goal 1	Goal 2	Goal 3	Feed-back 1	Feed-back 2	Feed-back 3
Goal 1	Property 1	X	X	X									
	Property 2	X	X	X									
	Property 3	X	X	X									
Goal 2	Property 4				X								
	Property 5					X	X						
	Property 6										X	X	X
Goal 3	Property 7							X	X	X			
	Property 8							X	X	X			
	Property 9							X	X	X			
Goal 4	Property 10							X	X	X			
	Property 11							X	X	X			

Table 10: Concept Matrix Mapping Evaluation Goals to Episodes for Thomas Becker

		Episode 1			Episode 2			Episode 3			Episode 4		
		Habit 1	Habit 2	Habit 3	Form Completion	Modification 1	Modification 2	Goal 1	Goal 2	Goal 3	Feed-back 1	Feed-back 2	Feed-back 3
Goal 1	Property 1	X	X	X									
	Property 2	X	X	X									
	Property 3	X	X	X									
Goal 2	Property 4				X								
	Property 5					X	X						
	Property 6										X	X	X
Goal 3	Property 7							X	X	X			
	Property 8							X	X	X			
	Property 9												
Goal 4	Property 10							X	X	X			
	Property 11							X	X				

As can be seen from Table 8, Table 9, and Table 10, all properties under Goal 1 and Goal 2 were achieved for the three personas. The properties under Goal 1 and Goal 2 are more oriented toward testing whether the system modules function properly, that is, Strategy 1: Technical Risk & Efficacy. From a functional perspective, habit classification, context extraction, ontology mapping, form completion, form modification, as well as recommendation adaptation based on user feedback, can be performed.

With regard to Goal 3, Property 7 source-based explanation of recommendation context and Property 8 source-based explanation of the recommendation itself were achieved. However, with regard to Property 9 consistency between recommendation style and user preference, the test of this property is more oriented toward Strategy 2: Human Risk & Effectiveness, that is, it focuses more on evaluating the utility of the system for users. Whether Property 9 can be achieved depends on the user's preferences. For example, the preferences of the persona Anna Müller include "step-by-step", but this preference was not realized in the recommendations generated for her three goals. It can be clearly seen that "step-by-step" was eventually passed into the M3.4 Habit Recommendation Generator. However, possibly because the current knowledge base does not contain step-by-step examples, this preference was not realized in any of the final recommendations. In addition, the preferences of the persona Thomas Becker include "fast results", but this preference was also not realized in the recommendations generated for his three goals. This shows that the realization of Property 9 depends not only on the system, but also on the degree of matching with the knowledge base, and may even be related to the LLM used for generating recommendations. Therefore, Goal 3 was only partially achieved.

With regard to Goal 4, Property 10 retrieval of knowledge-base evidence relevant to the recommendation task was achieved for all three personas. The reliability of retrieval is based on mature RAG theory, the embedding model BGE-M3, and the vector database Milvus, among others (Wang et al., 2021a; Chen et al., 2025). Here, the evaluation focuses on whether the retrieved evidence corresponds to the predefined gold evidence. For Property 11 reasonable evidence-grounded recommendations, most recommendations were able to provide reasonable recommendations based on evidence. However, a small number of hallucination-like phenomena still occurred. For example, the form input of the persona Anna Müller included "only 5–10 minutes per day for behavior change", but the recommended behavior was "Set a timer to stand up and do a brief 5-minute stretch or walk around the office each hour." Although the information "only 5–10 minutes per day for behavior change" entered the M3.4 Habit Recommendation Generator, the system still generated a recommendation that was inconsistent with this limitation. In addition, the prompt of the M3.4 Habit Recommendation Generator explicitly states that specific numbers must not be generated unless they are explicitly supported

by literature, the user profile, or user habits. Nevertheless, the persona Thomas Becker still received the recommendation “Take a short, gentle walk (about 10–15 minutes)”, even though there was no evidence supporting this specific duration. Therefore, Goal 4 was also only partially achieved.

5 Discussion

The preliminary evaluation indicates that the proposed Habit Recommender System is technically feasible as a prototype. The results of the Evaluation stage show that the main technical workflows, including habit modeling, user profile completion, feedback-based adaptation, knowledge base retrieval, and recommendation generation, could be successfully executed in the evaluated persona-based episodes. This suggests that combining RAG with users' behavior change goals, habits, user profile information, and feedback to generate personalized health recommendations represents a feasible design direction for AI-supported HRSs.

However, a technically executable prototype does not automatically guarantee that the generated recommendations fully match user preferences or are completely evidence-based. The main limitations became visible at the level of recommendation generation. More specifically, users' style preferences could not always be reflected in the final recommendations, and in some cases, the system still generated concrete numerical details that were not supported by the retrieved evidence. This indicates that prompt-level constraints alone are insufficient to fully control LLM-based recommendation generation. The quality of the final recommendations may depend not only on the system architecture, but also on the coverage of the knowledge base and the performance of the LLM.

From a DSR perspective, the main contribution of this thesis lies in the structured derivation and implementation of design knowledge for AI-supported HRSs. The identified meta-requirements, DPs, and DDs provide a traceable path from the problems identified in existing HRS research to the concrete implementation of the artifact. The Habit Recommender System is therefore not only a technical prototype, but also an instantiation of reusable design knowledge.

However, several limitations remain. First, at the evaluation level, the evaluation was preliminary and persona-based, and did not involve real users. The use of the same knowledge base for all three personas improved the comparability of the evaluation results, but also reduced the realism of the evaluation setting. Second, at the technical level, although RAG can mitigate hallucinations to some extent, hallucination-like outputs and unsupported details cannot be fully excluded by merely modifying prompts.

6 Conclusion and Future Work

This thesis designed and implemented a RAG-enhanced and ontology-informed Habit Recommender System to support personalized habit recommendations in the context of NCD prevention. The research followed a design science research approach and developed the artifact through the systematic derivation of meta-requirements, design principles, design decisions, a conceptual design, and a prototype implementation.

With regard to RQ1, four meta-requirements were identified for AI-supported HRSs: unified modeling and integration, dynamic personalization and continuous updating, interpretability and transparency, and trustworthiness and robustness. With regard to RQ2, these meta-requirements were translated into corresponding design principles, including semantic unification, context-rich user modeling, feedback-driven adaptivity, inspectable recommendation logic, and robust recommendation design. With regard to RQ3, these design principles were further operationalized into concrete design decisions and implemented in a modular prototype.

With regard to RQ4, the meta-requirements were not evaluated directly in their abstract form. Instead, they were operationalized into feasible evaluation goals and evaluation properties. The preliminary evaluation showed that the artifact could successfully execute the main technical functions in the evaluated persona-based episodes. However, the system did not always generate recommendations that fully reflected user-specific style preferences, and in a small number of cases, unsupported concrete numerical details were generated.

Based on these findings, several directions for future work can be derived. First, future evaluations should involve real users or domain experts in order to assess the artifact under more realistic conditions ([Qatar Computing Research Institute et al., 2022](#)). Second, different LLMs should be tested to examine how the choice of LLM influences recommendation outputs. In addition, fine-tuning or AI agent-based approaches could be explored to enhance the recommendation capabilities of LLMs ([Zhao et al., 2024](#); [Wu et al., 2024](#); [Wang et al., 2024](#)). Third, future work could develop a habit-specific ontology and explore the automatic collection of users' habits and contextual information through interaction with sensor-based data sources ([Mojarad et al., 2020b,a](#)).

List of Figures

1	DSRM-Based Research Process and Outputs	18
2	Traceability Mapping from Literature-Informed Problem Identification to Meta-Requirements, DPs, DDs, and the Artifact (based on Peffers et al. (2007); Koppenhagen et al. (2012))	18
3	Identified problems and specific research problem in AI-supported HRSs	25
4	From Identified Problems to Meta-Requirements	26
5	From Meta-Requirements to DPs	30
6	From Design Principle 1.1 to DDs	31
7	LLM performance (Kappa) on the habit classification task in a zero-shot setting for the English and German datasets	33
8	Mean accuracy (embedding-based) of different models in the zero-shot context extraction task	33
9	From Design Principle 1.2 to DDs	37
10	From Design Principle 2 to DDs	39
11	From Design Principle 3 to DDs	41
12	From Design Principle 4 to DDs	44
13	Conceptual Model of the Habit Recommender System	46
14	Workflow 1: Habitual Structured Collection	48
15	Workflow 2: User Forms Completion	49
16	Workflow 3: Habit Recommendation	50
17	User interfaces supporting Workflow 1, including habit donation and habit management	53
18	User interface supporting Workflow 2 for health profile management	54
19	User interfaces supporting Workflow 3, including recommendation generation and recommendation management	54
20	Three evaluation-oriented personas and their color-coded mapping to the meta-requirements	59
21	Demonstration example of the recommendation result for Persona 1	62
22	Adapted from the FEDS evaluation design process (Venable et al., 2016) and applied to the preliminary evaluation of the Habit Recommender System	63

List of Tables

1	Summary of Related Studies on AI-Supported HRSs	6
2	Overview of Included Studies, Modeled Objects, and Main Approaches	15
3	Candidate Ontologies Potentially Relevant for Habit Modeling Identified in the Previous Research Seminar and in the Current Non-systematic Literature Review on AI-supported HRSs	34
4	Example of mapping habit-related contextual elements to the BCIO	36
5	Mapping between system modules and corresponding DDs	47
6	Mapping between system modules, API endpoints, and architectural layers . . .	53
7	Overview of the Three Personas and the Literature Used for Their Construction	60
8	Concept Matrix Mapping Evaluation Goals to Episodes for Anna Müller	67
9	Concept Matrix Mapping Evaluation Goals to Episodes for Laura Schneider . .	68
10	Concept Matrix Mapping Evaluation Goals to Episodes for Thomas Becker . .	69

List of Listings

1	Example directory structure of the local knowledge base	56
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Declaration of Authorship

I hereby certify that this thesis has been composed by me and is based on my own work, unless stated otherwise. No other person's work has been used without due acknowledgement in this thesis. All references and verbatim extracts have been quoted, and all sources of information, including graphs and data sets, have been specifically acknowledged.

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